



**Verified Carbon
Standard**

DONGXING WASTE INCINERATION PLANT LEACHATE BIOGAS HARMLESS TREATMENT AND COMPREHENSIVE UTILIZATION PROJECT



New Power
百川畅银

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project (hereafter referred to as the project) is located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P.R. China. The project is constructed and operated by Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd.

Before the implementation of this project, Zhengzhou (East) environmental Protection Energy Project invested by Zhengzhou Dongxing Environmental Protection Energy Co., Ltd. incinerates domestic waste with a daily treatment capacity of 4,000t /d. Leachate will be produced during the storage and fermentation process in the garbage bin. The supporting treatment system of the factory is "pretreatment +UASB (upflow anaerobic sludge blanket) + secondary A/O-MBR aerobic system + nanofiltration NF+STRO" treatment process. A large amount of biogas is produced in the anaerobic treatment section of UASB. The Zhengzhou (East) environmental Protection Energy Project started to construct in 2018¹ and started operation in 07/2019², Because it was difficult to collect and treat biogas, the biogas generated by the UASB was emitted to the atmosphere directly without any recovery and utilization. The biogas was being vented to the atmosphere in 07/2019 when the Zhengzhou (East) environmental Protection Energy Project started operation.

This project collects the biogas generated by the anaerobic treatment section of leachate to generate electricity and connect to the grid, which can effectively utilize biogas resources.

The project started to construct on 15/09/2021 and started operation on 21/01/2022. The project has a total designed capacity 3MW (3*1MW) and combusts the leachate biogas, which contains nearly 65% of methane, to generate electricity and export it to the Central China Power Grid (CCPG). Total electricity supplied to the grid in the crediting period will be 118,470MWh. VCUs will be claimed from the methane destroyed and the grid electricity displaced. The estimated average GHG emission reductions of the project are 66,310tCO₂e

¹ <https://www.zzgyjt.cn/news/881.html>

² <https://www.zzgyjt.cn/news/861.html>

and the total GHG emission reductions and removals are 663,100tCO₂e during the fixed 10 years crediting period.

Scenario existing prior to the implementation of the project (the same as baseline scenario):

Leachate biogas from Dongxing Waste Incineration Plant wastewater treatment is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the CCPG, which is dominated by fossil fuel based power plants.

1.2 Sectoral Scope and Project Type

Sector scope 1: energy industries (renewable-/non-renewable sources)

Sector scope 13: waste handling and disposal

The project is not a grouped project.

1.3 Project Eligibility

The project is a leachate biogas power generation project, utilizing leachate biogas, which consists mainly of methane, for electricity generation, which is eligible under the scope of the VCS program.

1.4 Project Design

The project has been designed to include a single installation of an activity.

Eligibility Criteria

The project is not a grouped project. Thus, this section is not applicable.

1.5 Project Proponent

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Project Manager

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1.6 Other Entities Involved in the Project

Organization name	Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd
Role in the project	Project Operation Entity
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Title	Project Manager
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1.7 Ownership

Zhongmu Branch of Henan BCCY Environmental Energy Co., Ltd. is the sole project owner of the project. The approval of Feasibility Study Report (FSR) and Environmental Impact Assessment (EIA) and the Business License of the project owner are the evidences for legislative right. Henan BCCY Environmental Energy Co., Ltd. is the mother company of Zhongmu Branch of Henan BCCY Environmental Energy Co., Ltd., and Henan BCCY Environmental Energy Co., Ltd. in Zhongmu project City authorized Henan BCCY Environmental Energy Co., Ltd. as the Project Proponent.

1.8 Project Start Date

21/01/2022 (commercial operation started date)

1.9 Project Crediting Period

The project crediting period runs from 21/01/2022 through 20/01/2032 for 10 years fixed.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
21/01/2022-31/12/2022	62,677
2023	66,310
2024	66,310
2025	66,310
2026	66,310
2027	66,310
2028	66,310
2029	66,310
2030	66,310
2031	66,310
01/01/2032-20/01/2032	3,633
Total estimated ERs	663,100
Total number of crediting years	10
Average annual ERs	66,310

1.11 Description of the Project Activity

GHG emission reductions will be achieved through combustion of the recovered methane gas by gas engines, which would be otherwise emitted to the atmosphere, and the generation of electricity from leachate biogas, which is supplied by the CCPG prior to the implementation of the project.

The leachate biogas collection system and treatment system will emit the biogas to the atmosphere directly. The electricity generation and grid connection system will generate electricity to the CCPG.

Description of the technology in the project

The project consists in leachate biogas collection, transmission and treatment system, with subsequent electricity generation and grid connection system.

Leachate biogas collection system

Leachate biogas is extracted from UASB anaerobic biochemical system by the top set gas pipe collection into the storage tank. The pipeline is connected to the biogas storage tank through Roots fan to extract and transport biogas into the biogas pretreatment unit. Flow rate of the leachate biogas is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

Leachate biogas treatment system

Prior to electricity generation, leachate biogas is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

3 gas engines of 1MW rated power each are fed with the leachate biogas and generate electricity, which is then exported to the grid.

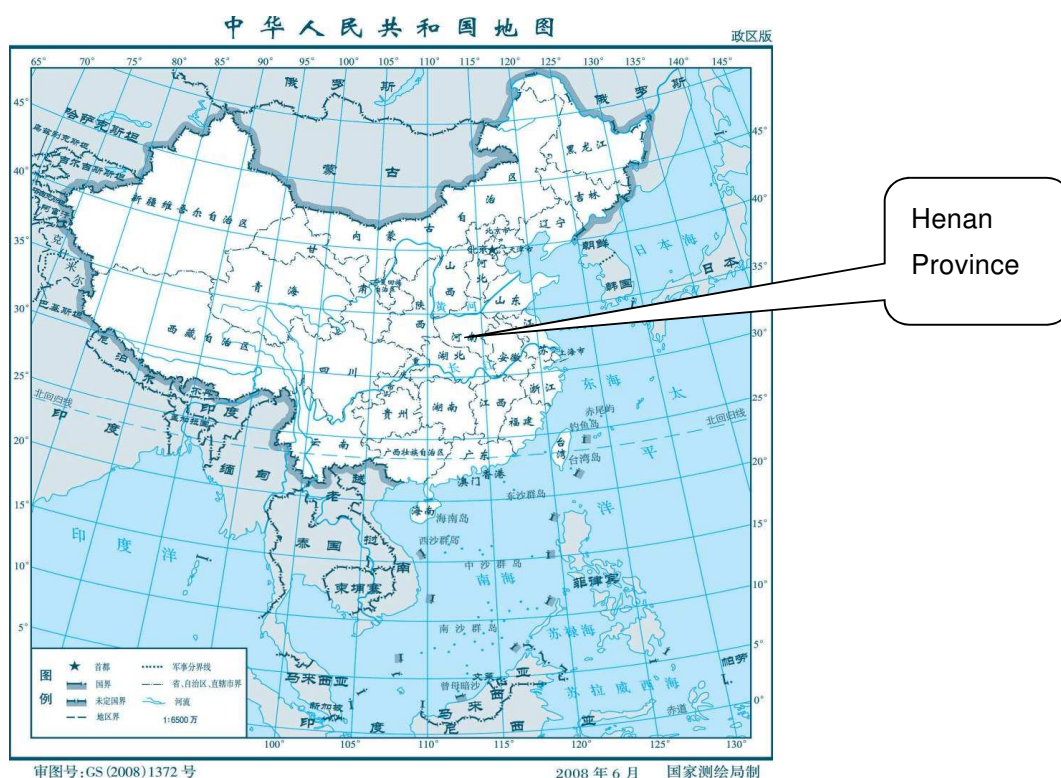
Table 1.1 Equipment technical parameters

Main Equipment	Parameter	Value
Gas engine	Type	J320GS
	Manufacturer	INNIO Jenbacher GmbH & Co OG
	Rated capacity	1000 kW

	Rated rotation speed	1500 r/min
	Lifetime	20 year

1.12 Project Location

The project is located in Dongxing Waste Incineration Plant, Zhengyoumo Village, Zhengan Town, Zhongmu County, Zhengzhou City, Henan Province P.R. China. The geographical coordinates of the project site are longitude 113.57°E and latitude 34.37°N. Figure 1.1 shows the location of the project.



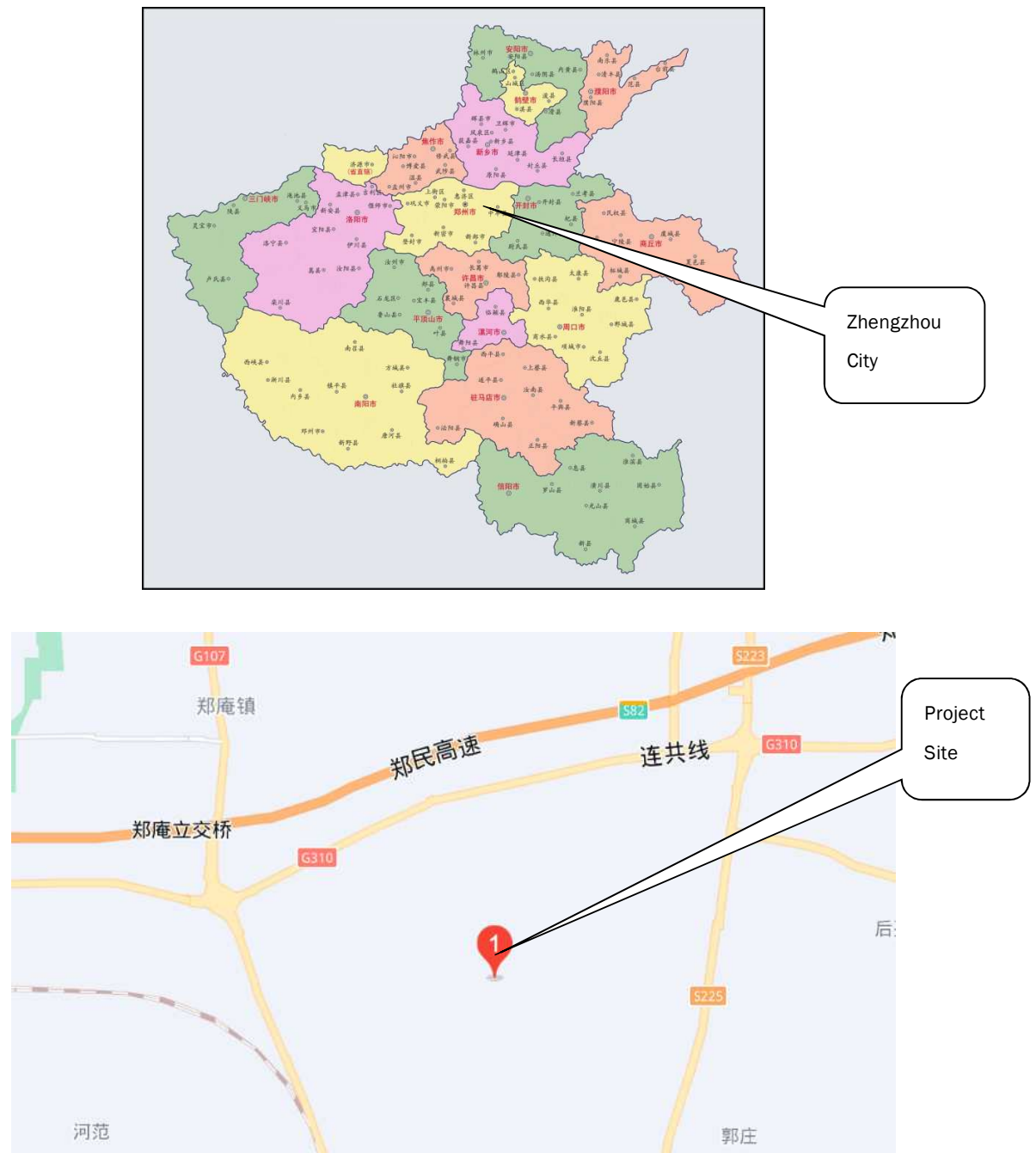


Figure 1.1 Location of the project

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity (the same as baseline scenario)

- Leachate biogas from Dongxing Waste Incineration Plant is released into the atmosphere.
- Equivalent electricity generated by the project is supplied by the CCPG, which is dominated by fossil fuel based power plants.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Chinese relevant laws and regulations. Mainly include:

1. Renewable Energy Law of the People's Republic of China;
2. National Action plan for the collection and Utilization of municipal leachate biogas;
3. Catalogue for the Guidance of Industrial Structure Adjustment (2011 version).

The project obtained the approval letters from local governmental authorities: Investment Supervisory Department, as well as Environment Protection Agency (EPA). The two approvals well demonstrate that local governments permit the construction of the project. Hence, the project is in compliance with laws, status and other regulatory frameworks.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is not registered or is seeking registration under other GHG Programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project didn't reduce GHG emissions from activities that are included in emissions trading program or any other mechanism that includes GHG allowance trading, and include details about any such programs or mechanisms.

1.16.2 Other Forms of Environmental Credit

The project hasn't sought or received another form of GHG-related environmental credits, including renewable energy certificates.

The Green Energy Certificate (GEC) is a voluntary renewable energy trading system that has been implemented in China since 2017. Only wind power (onshore wind power) and photovoltaic power generation projects (excluding distributed photovoltaic projects) listed in the national renewable energy price additional fund subsidy list will be taken as the verification object. The project does not meet its conditions.

1.17 Sustainable Development Contributions

The project is located in Zhengyoumo Village, Zhengnan Town, Zhongmu County, Zhengzhou City, Henan Province, P.R. China. The project captures leachate biogas and generates electricity, which is fed into the CCPG, displacing energy that would otherwise be generated by fossil fuels.

The primary objective of the project is to avoid greenhouse gases emission through leachate biogas capture, purification, and injection in a grid connection system, while contributing to the environmental, social, and economic sustainability by minimizing global climate changes and local air pollution. Additional benefits include the increase of employment opportunities through full-time and permanent positions, etc.

The Project will contribute to sustainable development in the following ways:

- Reduce CH₄ emission. The project will recover and destroy leachate biogas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030;
- Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in Central China Power Grid and further help to achieve the national action to promote renewable energy development;
- Improve the operational safety of the waste incineration plant. The project reduces the potential safety hazards of leachate biogas explosion by recycling leachate biogas and improves the safety of waste incineration plant operation;
- The project will also provide new jobs opportunities and increase tax revenue, which will have a positive effect on the local economy.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as not AFOLO project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No further information available.

2 SAFEGUARDS

2.1 No Net Harm

The Environment Impact Assessment (EIA) was compiled by Henan Langtian Environmental Technology Co., Ltd. in September, 2021 and approved by Ecological environment Bureau of Zhengzhou County on 15/10/2021. Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project development. The environmental impacts, treatment and effect arising from the Project during operation has been analysed in section 4 of this report below. And no net harm has been detected.

Meanwhile, the implementation of the project will improve local socio-economic development and contribute to the sustainable development as described in section 1 of this report above.

In conclusion, the project has no negative impacts on local environment and socio-economy. No net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

The project owner distributed questionnaires to local residents who may be impacted by the project in order to collect advice for the project. The aim of these questionnaires was to collect

opinions concerning the influence the project would have on the local society, environment, economy, daily life etc.

The following questions are from the questionnaires and stakeholder consultancy meeting:

1. What do you think about the environment of surrounding areas?
2. Do you support the construction of this leachate biogas power station?
3. Do you think the implementation of the project will cause positive effect on living of local residents?
4. Do you think the implementation of the project will cause negative effect?

The stakeholders were informed about the stakeholder meeting through posters on 10/11/2021. In the bulletin, the company invited the potential stakeholders to get to know their opinions and/or suggestions about the implementation of the project and VCS application of the project. The stakeholder consultancy meeting for the parties interested in the project was organized at Dongxing Waste Incineration Plant on 25/11/2021 to collect opinions from all stakeholders, such as representatives of local residents, Dongxing Waste Incineration Plant, the project owner and so on. The project owner appointed one person to make a meeting minute.

30 questionnaires were distributed, and 30 questionnaires were returned. The age range of questioned people was 23-65 years old.

- About 30% of questioned people think the environment of surrounding areas is bad, and others think it is acceptable and good.
- 100% of questioned people support the construction of the project.
- 96% and 83% of questioned people think the implementation of the project will mitigate air pollution and lack of electricity, respectively.
- 60% of questioned people think the noise of implementation of the project will affect the living of local residents surround.

There are 15 stakeholders attended the stakeholder consultation meeting and the project owner had taken meeting minute.

During the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project owner will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried

out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures.

Expect for introducing the project information, grievance mechanism was also explained during the local stakeholder consultation meeting. A grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project Participant will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Participant will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible.

As a whole, the impacts of construction of the project are basically positive. All stakeholders were pleased with the development of the project. The project would actually facilitate the development of the local economy and increase the income of local residents.

2.3 Environmental Impact

In September 2021, an environmental impact assessment (EIA) was completed in accordance with Chinese regulation by Henan Langtian Environmental Technology Co., Ltd. and was approved by Ecological environment Bureau of Zhengzhou County on 15/10/2021. The objective of this EIA was to identify the effects of the project activities on both the biophysical components of the environment and socio-economical aspects of local community and to provide measures and procedures to mitigate the possible effects.

The outcome of the EIA was favourable and the project was found to have no significant environmental impacts. The project not only reduces the danger caused by uncontrolled release of leachate biogas, but also reduces the pollution caused by the leachate biogas to the air, soil and water quality in the local area. From environmental protection perspectives, the project is in compliance with national industry policy, promoting sustainable development and utilisation of waste.

During construction phase, the raise dust, noise, waste water and solid wastes caused by project construction will be treated according to the measures in EIA report, and there will be no significant impact on the environment.

During the operation period, all the mitigation measures proposed by EIA report will be implemented and the following key aspects will be addressed:

Waste water

The wastewater produced by the project consists of biogas condensation waste water, circulating cooling waste water of generator set, and reclaimed waste water of soft water equipment. Condensation wastewater will be treated by leachate treatment system of Zhengzhou Dongxing Environmental Protection Energy Co., LTD., the sewage is reused and not discharged. Therefore, it can meet the standards of "The reuse of urban recycling water-Waste quality standard for industrial uses"(GB/T19923-2005).

Waste Gas

The exhaust gas produced by this project is mainly the exhaust gas of the generators. The concentration of the exhaust gas (mainly SO₂, NO_x and particulate matter) after combustion in gas-generator sets and processed by the SCR denitration system is discharged through the 15m high cylinder exhaust.

Noise pollution

According to the prediction results, the acoustic environment quality of the project equipment noise transmitted to the northwest boundary of project can meet the Class 2 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008). The acoustic environment quality of the project equipment noise transmitted to the southeast boundary of project can meet the Class 4 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008).

Solid waste

The solid waste of the project is mainly sulfur paste, waste oil, waste oil barrel and waste denitration catalyst. The sulfur paste is stored in the sulfur storage room and sold regularly. The waste oil, waste oil barrels and waste denitration catalyst are stored in the hazardous waste room of the factory, and regularly handed over to qualified units for recycling and processing. The construction standard of hazardous waste room shall meet the requirements of "Standard for pollution control on hazardous waste storage" (GB18597-2001).

As mentioned above, the project was considered not causing significant environmental impacts.

2.4 Public Comments

There are no comments received during the public comment period (29/11/2022 to 29/12/2022).

Reference: <https://registry.verra.org/app/projectDetail/VCS/3851>

2.5 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Title: AMS-III.H. Methane recovery in wastewater treatment --- Version 19.0

AMS-I.D. Grid connected renewable electricity generation---Version 18.0

Reference: <https://cdm.unfccc.int/methodologies/index.html>

The methodology also refers to the latest approved versions of the following tools:

“Tool to calculate the emission factor for an electricity system” (Version 07.0);

“Tool for the demonstration and assessment of additionality” (version 07.0.0);

“Demonstration of additionality of small-scale project activities” (version 13.1);

“Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (Version 3.0)

3.2 Applicability of Methodology

The selected methodology AMS-III.H. (version 19.0) and AMS-I.D. (version 18.0) are appropriate to leachate biogas project activities, where the baseline scenario is the atmospheric release of the leachate biogas, and that all or part of the electricity exported to the grid is the electricity generation in existing and/or new grid-connected power plants. In this case, the leachate biogas is released to atmosphere prior to the implementation of the project, and electricity generated by the project is exported to Central China Power Grid (CCPG).

The project fulfils the following applicability conditions of the methodology:

Applicability conditions for AMS-III.H.	Justifications
2.This methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; c) Introduction of biogas recovery and combustion to a sludge treatment system; d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment, to an untreated wastewater stream; f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	Applicable. This project collects the biogas generated by the anaerobic treatment section of leachate to generate electricity and connect to the grid. The simplified baseline methodology is applicable to this project activity because without the project activity, methane from the existing anaerobic treatment system would continue to be emitted into the atmosphere. As such, paragraph 2(f) of the methodology applies “Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery)”
3.In cases where baseline system is anaerobic lagoon the methodology is applicable if: (a) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or	Applicable. a) The baseline system is anaerobic lagoons with the minimum depth of 2 m; b) The average atmospheric temperature in the region is above 15°C from April to

through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (b) Ambient temperature above 15°C, at least during part of the year, on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	October³; c) The sludge is incinerated in the incineration system of this project. There was no sludge generation in the baseline scenario. Therefore, the project activity is applicable for this criterion.
4.The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: a) Thermal or mechanical, electrical energy generation directly; b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users; d) Hydrogen production; e) Use as fuel in transportation applications after upgrading.	Applicable. The recovered biogas will be utilized to generate electrical energy directly. As such, paragraph 4 (a) applies.
5.If the recovered biogas is used for project activities covered under paragraph 4(a), that	Applicable.

³ https://www.tianqi.com/qiwen/city_zhongmou/

component of the project activity can use a corresponding methodology under Type I.	Since the biogas captured will be used for electrical energy generation directly for this project activity, methodology under Type I should be used for the electricity generation components. Therefore, AMS.I.D is applied for the project activity.
6.For project activities covered under paragraph 4(b), if bottles with upgraded biogas are sold outside the project boundary, the end-use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period CO ₂ emissions avoided by the displacement of fossil fuel can be claimed under the corresponding Type I methodology, e.g. “AMS-I.C.: Thermal energy production with or without electricity”	Not applicable.
7.For project activities covered under paragraph 4(c)(i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	Not applicable.
8.For project activities covered under paragraph 4(c)(ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	Not applicable.
9.In particular, for the case of paragraph 4(b) and (c)(iii), the physical leakage during storage and transportation of upgraded biogas, as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant	Not applicable.

procedures in paragraph 18 of the appendix of “AMS-III.H.: Methane recovery in wastewater treatment” shall be followed in this regard.	
10.For project activities covered under paragraph 4(b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these exist) or, in the absence of national regulations, a minimum of 96 per cent (by volume).	Not applicable.
11.If the recovered is utilized for the production of hydrogen (project activities covered under paragraph 3(d)), that component of the project activity shall use the corresponding methodology “AMS-III.O.: Hydrogen production using methane extracted from biogas”.	Not applicable.
12.If the recovered biogas is used for project activities covered under paragraph 4(e), that component of the project activity shall use corresponding methodology “AMS-III.AQ.: Introduction of Bio-CNG in transportation applications”.	Not applicable.
13.New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the relevant requirements in the “General guidelines for SSC CDM methodologies”. In addition, the requirements for demonstrating the remaining lifetime of the equipment replaced, as described in the general guidelines shall be followed.	Not applicable.
14.The location of the wastewater treatment plant as well as the source generating the wastewater shall be uniquely defined and described in the PDD.	Applicable. The location of the wastewater treatment plants is uniquely defined in the factory site of Dongxing Waste Incineration Plant. The source generating the wastewater is

	from production water and domestic sewage from waste incineration plant, and the wastewater is being imported to the leachate treatment station treatment through channels.
15.Measures are limited to those that result in aggregate emissions reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity.	Applicable. The estimated annual ERs are less than the 60,000 tCO ₂ limit for Type III components. The project activity estimated annual ERs are 59,533 tCO ₂ for AMS-III.H.

Applicability conditions for AMS-I.D.	Justifications
4.This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant (s); (c) Involve a retrofit of (an) existing plant (s); (d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or; (e) Involve a replacement of (an) existing plant(s).	Applicable. The project installs a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project, which corresponds to point (a).
5.Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per	Not applicable. The project is not a hydro power plant.

definitions given in the project emissions section, is greater than 4W/m²; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m².	
6. If the new unit has both renewable and non-renewable components (e.g. a wind/ diesel unit), the eligibility limit of 15 MW for a small-scale VCS project activity applies only to the renewable component. If the new unit co-fires fossil fuel. The capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable. The project does not use non-renewable components nor co-fires fossil fuels. Anyway, the total installed capacity is below 15MW.
7. Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable. The project only involves electricity generation.
8. In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable. The project does not involve addition of renewable energy generation units at an existing renewable power generation facility.
9. In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.	Not applicable.
10. In the case of leachate biogas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed	Applicable. The project installed 3 engines to combust leachate biogas of Dongxing waste incineration plant, which mainly contains 65% methane.

under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-L.C. Thermal energy production with or without electricity shall be explored.	
11. In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	Not applicable.

Regarding to tool— “Tool to calculate the emission factor for an electricity system” (Version 07.0):

Applicable conditions	Justifications
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. This tool is applied to estimate the OM, BM and/or CM.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity	Applicable. The emission factor for the project electricity system is calculated for grid power plants only.

generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
In case of VCS projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not applicable. The project electricity system is located in China.
Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	Not applicable.

Regarding to tool— “Positive lists of technologies” (Version 04.0):

Applicable conditions	Justifications
The use of this methodological tool is not mandatory for the project participants of a VCS project activity or VCS POA for demonstrating their additionality.	Applicable.
This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable. This tool is applied in conjunction with small-scale methodology AMS-III.H. (version 19.0) .

Regarding to tool— “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (Version 03.0):

Applicable conditions	Justifications
5.If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power	Applicable. (a) Scenario A: Electricity consumption from the grid.

plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
6. This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Applicable. (a) Scenario I: Electricity is supplied to the Central China Power Grid (CCPG);

7. This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO ₂ emissions.	Not applicable.
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3.3 Project Boundary

The project boundary is the site of the project activity, Dongxing Waste Incineration Plant, where the leachate biogas is captured and used, and also includes all the power sources connected to the CCPG, which expands throughout Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province and Chongqing City.

The treatment systems not affected by the project activity, i.e. sections operating in the project scenario under the same operational conditions as in the baseline scenario. The water treatment system of the factory is "pretreatment +UASB+ secondary A/O-MBR aerobic system + nanofiltration NF+STRO" treatment process. The project adds biogas collection, transmission and treatment system, with subsequent electricity generation and grid connection system. The wastewater inflow and COD content, temperature, retention time, etc. are not affected by the biogas collection.

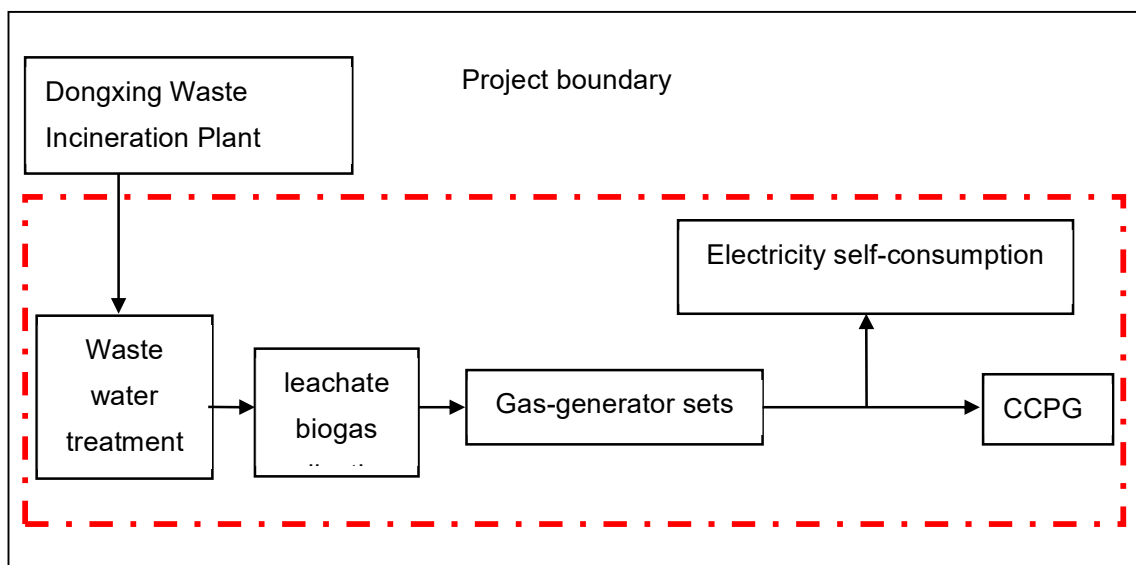


Figure 3.1 the Diagram of Project Boundary

Table 3.1 Summary of greenhouse gases and sources included in and excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from wastewater treatment system	CH ₄	Yes	The major source of emissions in the baseline.
		N ₂ O	No	Excluded for simplification.
		CO ₂	No	CO ₂ emissions from the decomposition of organic waste in the wastewater are not accounted for.
	Emission from electricity generation	CO ₂	Yes	The major source of emissions in the baseline.
		CH ₄	No	Excluded for simplification, this is conservative.
		N ₂ O	No	Excluded for simplification, this is conservative.
	Emissions from heat generation	CO ₂	No	No heat generation is included in the project.
		CH ₄	No	No thermal energy generation is included in the project.
		N ₂ O	No	No thermal energy generation is included in the project.
	Emissions from the use of natural gas	CO ₂	No	Excluded for simplification. This is conservative
		CH ₄	No	No supply of leachate biogas is included in the project.
		N ₂ O	No	Excluded for simplification. This is conservative
Project activity	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO ₂	No	Excluded because there is no fossil fuel consumption.
		CH ₄	No	Excluded because there is no fossil fuel consumption.
		N ₂ O	No	Excluded because there is no fossil fuel consumption.
	Emission from electricity consumption due to project activity	CO ₂	Yes	Main emission source.
		CH ₄	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.

Source	Gas	Included?	Justification/Explanation
	N ₂ O	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.
	CO ₂	No	Excluded because there is no flaring in the project.
	CH ₄	No	Excluded because there is no flaring in the project
	N ₂ O	No	Excluded because there is no flaring in the project
	CO ₂	No	CO ₂ emissions from the decomposition of organic waste in the wastewater are not accounted for.
	CH ₄	Yes	CH ₄ is the major component in the biogas produced during anaerobic wastewater treatment.
	N ₂ O	No	Excluded for simplification.

3.4 Baseline Scenario

Leachate biogas: In the absence of the project, Leachate biogas of Dongxing Waste Incineration Plant is left to decay within the project boundary, and methane is emitted to the atmosphere directly without any recovery and utilization.

Electricity: The project is a new grid-connected renewable power unit.

According to the section 5.3 of the methodology AMS-III.H. and 5.2 of methodology AMS-I.D., the baseline scenario is:

Leachate biogas from Dongxing Waste Incineration Plant is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the CCPG, which is dominated by fossil fuel based power plants.

3.5 Additionality

As per “Demonstration of additionality of small-scale project activities” Version 13.1, additionality of the project shall be demonstrated by providing an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- (a) Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions;
- (b) Technological barrier: a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions;
- (c) Barrier due to prevailing practice: prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions;
- (d) Other barriers: without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

In line with the above guidance, the additionality is demonstrated using option (a) Investment barrier. The tool available in “Tool for the demonstration and assessment of additionality, version 07.0.0” is used to conduct the investment analysis and determine appropriate analytical methods.

Option I. Apply simple cost analysis

Applicable to situations where the proposed project does not generate financial or economic benefits other than those related to emission reduction mechanisms. In addition to VCUs revenue, this project also includes electricity sales revenue, so Option I is not applicable to this project.

Option II. Apply investment comparison analysis

Suitable for alternative projects as well as investment projects. Providing equal power supply from Central China Power Grid is not a specific investment project, and the project participants do not participate in investment related to the construction of the power grid. Therefore, Option II is not applicable to this project.

Option III. Apply benchmark analysis

Applicable to situations where alternative projects are not investment projects.

Therefore, the benchmark analysis (Option III) option has been applied. This project adopts a benchmark rate of return analysis to analyze the investment obstacles of the project. According to the "Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects"⁴, the post tax benchmark rate of return for the full investment of landfill gas power generation projects is 8%.

Based on the Feasibility Study Report of the proposed project, basic parameters for calculation of financial indicators are as follows:

Table 3.2 Main parameters used in the calculation of the after-tax project IRR

Parameters	Unit	Value	Source
Total static investment	Million RMB	17.53	FSR
Installed capacity	MW	3	FSR
Net electricity supplied to the grid	MWh	11,847	FSR
Electricity Tariff (with VAT)	RMB/kWh	0.3779	FSR
VAT for electricity	%	17	FSR
Urban maintenance and construction tax	%	7	FSR
Education additional tax	%	3	FSR
Income tax	%	25	FSR
Lifetime	Year	10	FSR
Residual value rate of fixed assets	%	5	FSR
Annual operation & maintenance costs in a normal year	Million RMB	2.339	FSR
Expected VCUs price	RMB /tCO ₂ e	24	Assumption
Plant load factor (PLF)	%	45.08 ⁵	Calculated

⁴ "Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects", issued by State Power Corporation on 10/09/2002.

⁵ $PLF = 11,847 / (3 \times 8,760) = 45.08\%$.

Comment: The FSR refers only to the biogas capture and electricity generation investment and O&M costs.

The benchmark analysis compares the after-tax project IRR with the benchmark. The main results of the financial analysis are presented in Table 3.2, where the project IRR with and without VCS revenues are compared to the benchmark.

Table 3.3 Comparison of calculated Project IRR and the benchmark

Project IRR	Benchmark	Project IRR
Without VCS	Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects	With VCS
3.40%	8%	16.64%

It is clear from the table above that without VCS revenue the project IRR is far below the benchmark, in others words, it is financially unattractive, while with VCS revenue the project IRR is negative.

According to the financial analysis, after the implementation of the project with the expected input-output effect, the financial internal rate of return of all investment of the project is 3.40%. The economic benefit of the project is poor and lower than benchmark 8%. After the transaction of VCS project is included, the financial internal rate of return after income tax increases to 16.64%. The financial profitability of the project is good, and the income target can be reached. Therefore, after applying for VCS project, the project income can be increased.

A sensitivity analysis is conducted with the purpose of checking whether the conclusion regarding the financial viability of the proposed project is sound and tenable with those reasonable variations in the assumptions. The investment analysis provides a valid argument in favor of additionality only if it consistently supports (for a realistic range of assumptions) the conclusion that the project activity is unlikely to be the most financially attractive or is unlikely to be financially attractive. The following financial parameters were identified as the main variable factors for sensitivity analysis of financial attractiveness:

- Total static investment
- Electricity tariff
- Net electricity supplied to the grid
- Annual O&M cost

Table 3.3 presents the results of the sensitivity analysis for the four main parameters. Financial analyses were performed to assess the impact on the project profitability by altering each of these parameters by $\pm 10\%$ as conservative consideration. The impact on the project IRR is as follows:

Table 3.4 sensitivity of Project IRR to different financial parameter

	-10%	-5%	0%	5%	10%
Total static investment	5.01%	4.18%	3.40%	2.65%	1.91%
Electricity tariff	-0.21%	1.65%	3.40%	4.89%	6.32%
Net electricity supplied to the grid	-0.21%	1.65%	3.40%	4.89%	6.32%
Annual O&M cost	3.56%	3.49%	3.40%	3.26%	3.13%

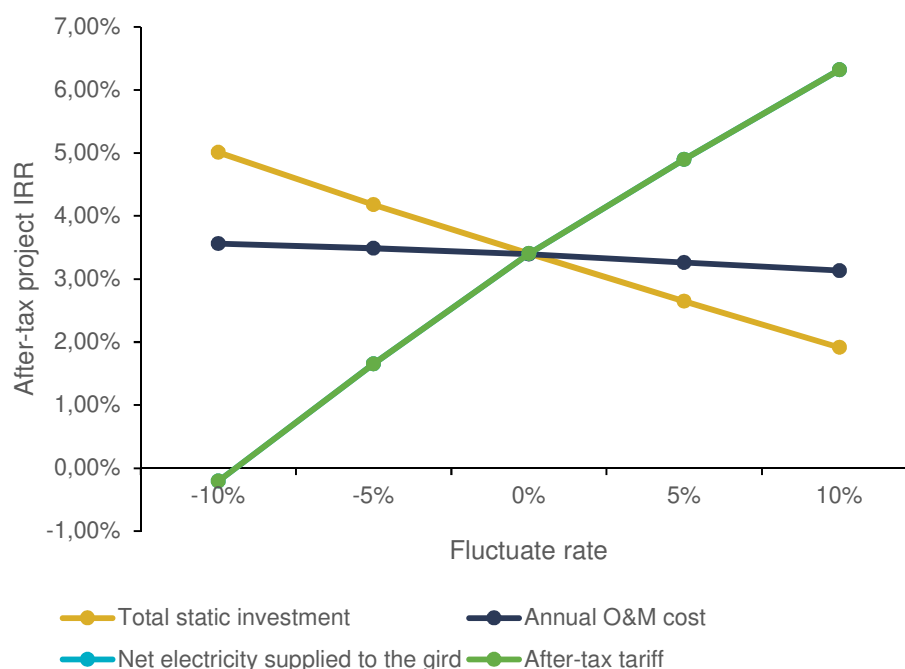


Figure 3.2 Sensitivity analysis of after-tax project IRR with changes of key parameters

Total static investment: With a decrease in the total static investment by 10%, the after-tax Project IRR is 5.01% still less than 8.0%. Even when the total static investment reduced by 24.95% (decrease to 13,160,000 RMB), the after-tax project IRR will reach benchmark. According to the fixed asset investment price index of National Bureau of Statistics⁶, the fixed

⁶ <https://data.stats.gov.cn/easyquery.htm?cn=C01&zb=A0901&sj=2022>

asset investment price index from 2017 to 2019 was respectively is 101.0, 101.2 and 103.5 and It's always growing. Therefore, the static total investment of the project cannot be reduced by 24.95%. According to the audit of financial accounts, the total cost has been 16,887,111.78 RMB, which exceeds the sensitive point of the sensitivity analysis of the project. Therefore, it is not possible to increase the economic attractiveness of the project by reducing the total static investment.

Annual O&M cost: With a decrease in the annual O&M cost by 10%, the after-tax Project IRR would only rise to 3.56% which is much lower than the benchmark of 8.0%. When the O&M cost decreases to 0 RMB, the IRR will not reach benchmark, however such great decrease is unlikely happen due to the increasing CPI in China⁷.

Electricity tariff: With an increase in the electricity tariff by 10%, the after-tax Project IRR of the project is 6.32%, less than 8.0%. When the electricity tariff increases to 0.4388 RMB/kWh the after-tax Project IRR of the project can reach benchmark of 8.0%. However, the approval of electricity tariff of a power generation project is set by the government⁸. The actual electricity tariff of the project is 0.3779RMB/kWh⁹ (As indicated in the electricity purchase and sale contract) , as per the approval of electricity tariff. Therefore, it cannot be expected that the after-tax Project IRR will improve due to an increase in the electricity tariff.

Net electricity supplied to the grid: With an increase in net electricity supplied to the grid by 10%, the after-tax Project IRR of the project is 6.32%, less than 8.0%. When the net electricity supplied to the grid increases by 16.11%, the after-tax Project IRR of the project can reach benchmark of 8.0%. Even when the generators will supply all generated electricity to the grid (Rate of electricity consumption by plant is assumed to be 0%) during the crediting period, the annual average electricity supplied to the grid will only reach 11,847¹⁰ MWh

⁷ <http://sousuo.www.gov.cn/sousuo/search.shtml?code=17da70961a7&dataTypeld=107&searchWord=CPI>

⁸ http://www.gov.cn/ztl/2006-01/20/content_165910.htm

⁹ 《Notice of Henan Provincial Development and Reform Commission on Matters related to Reasonable Adjustment of Electricity Price Structure》 [2017]707

¹⁰ According to the invoice for 2022, the net electricity supplied to the grid of 2022 is 8,975 MWh. The difference between 8,975 MWh and 8885 MWh ($11,847/12 \times 9 = 8,885$) is within reasonable range, since power generator 2# was started operation on 21/01/2022, power generator 1# was started operation on 26/04/2022 and 3# was started operation on 28/04/2022. So the annual net electricity supplied to the grid of 11,847 MWh is reasonable.

(according to EIA). Moreover, the net electricity supplied to the grid is calculated based on captured biogas, which is estimated conservatively. Therefore, it is very unlikely for the project to become commercially attractive through an increase of the electricity generation.

In conclusion, the above sensitivity analysis provides valid argument that the financial attractiveness of the proposed project is robust to reasonable variations in the critical assumptions, and consistently supports that without VCUs revenue, the proposed project is not financially attractive.

3.6 Methodology Deviations

No methodology deviation is applied in the project.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The project utilizes the leachate biogas for generation to substitute the equivalent electricity supplied by the grid, resulting in CH₄ and CO₂ emissions, which will be calculated as follows according to the methodology AMS-III.H. and AMS-I.D.

Baseline emissions associated with wastewater treatment:

$$BE_{y,1} = BE_{power,y} + BE_{ww,treatment,y} + BE_{s,treatment,y} + BE_{ww,discharge,y} + BE_{s,final,y}$$

Equation (1)

Where:

$BE_{y,1}$ = Baseline emissions associated with wastewater treatment in year y (t CO₂e)

$BE_{power,y}$	= Baseline emissions from electricity or fuel consumption in year y (t CO ₂ e)
$BE_{ww,treatment,y}$	= Baseline emissions of the wastewater treatment systems affected by the project activity in year y (t CO ₂ e)
$BE_{s,treatment,y}$	= Baseline emissions of the sludge treatment systems affected by the project activity in year y (t CO ₂ e)
$BE_{ww,discharge,y}$	= Baseline methane emissions from degradable organic carbon in treated wastewater discharged into sea/river/lake in year y (t CO ₂ e). The value of this term is zero for the case 1(b)
$BE_{s,final,y}$	= Baseline methane emissions from anaerobic decay of the final sludge produced in year y (t CO ₂ e). If the sludge is controlled combusted, disposed in a landfill with biogas recovery, or used for soil application in the baseline scenario, this term shall be neglected

There is no electricity consumption. Therefore, $BE_{power,y}$ is not applicable .

In the baseline scenario, treated wastewater is used for factory reusing and is not discharged into sea/lake/river, therefore methane emissions from degradable organic carbon in treated wastewater discharged ($BE_{ww, discharge, y}$) in e.g. a river, sea or lake in the baseline situation is assumed to be zero. Therefore, $BE_{ww, discharge, y} = 0$

The sludge is controlled combusted by sent to incinerator. Therefore, $BE_{s,final,y}$ and $BE_{s,treatment,y}$ is not applicable to calculations.

Therefore, the baseline emissions equation is simplified to:

$$BE_{y,1} = BE_{ww,treatment,y} \quad \text{Equation (2)}$$

Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline plant:

$$BE_{ww,treatment,y} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH4} \quad \text{Equation (3)}$$

Where:

$Q_{ww,i,y}$	= Volume of wastewater treated in baseline wastewater treatment system i in year y (m ³).
$COD_{inflow,i,y}$	= Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y (t/m ³). Average value may be used through sampling with the confidence/precision level 90/10
$\eta_{COD,BL,i}$	= COD removal efficiency of the baseline treatment system i, determined as per the paragraphs 35,36 or 37

$MCF_{ww,treatment,BL,i}$	=	Methane correction factor for baseline wastewater treatment systems i
$B_{o,ww}$	=	Methane producing capacity of the wastewater (IPCC value of 0.25 kg CH ₄ /kg COD)
UF_{BL}	=	Model correction factor to account for model uncertainties (0.89) ¹¹
GWP_{CH_4}	=	Global Warming Potential for methane

According to the EIA, the Volume of treated wastewater is 859.2 m³/d.

$$Q_{ww,i,y} = 859.2 \times 365 = 313,608 \text{ m}^3/\text{yr}$$

The Methane Correction Factor (MCF) shall be determined based on the following table:

Table 4.1. IPCC default values 7 for Methane Correction Factor (MCF)

Type of wastewater treatment and discharge pathway or system	MCF value
Discharge of wastewater to sea, river or lake	0.1
Land application	0.1
Aerobic treatment, well managed	0.0
Aerobic treatment, poorly managed or overloaded	0.3
Anaerobic digester for sludge without methane recovery	0.8
Anaerobic reactor without methane recovery	0.8
Anaerobic shallow lagoon (depth less than 2 metres)	0.2
Anaerobic deep lagoon (depth more than 2 metres)	0.8
Septic system	0.5
Land application ^(a)	0.1

$$MCF_{ww,treatment,BL,i} = 0.8$$

Baseline emissions associated with electricity generation ($BE_{EC,y}$)

$$BE_{y,2} = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (4)}$$

$BE_{y,2}$ = Baseline emissions associated with electricity generation in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh)

¹¹ Reference: FCCC/SBSTA/2003/10/Add.2, page 25.

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad \text{Equation (5)}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y(MWh)

Calculation of $EF_{grid,y}$

According to the AMS-III.G., $EF_{grid,y}$ is calculated using the “Tool to calculate the emission factor for an electricity system”(EF_{grid,y}=EF_{grid,CM,y}).

Project participants shall apply the following six steps:

- Step 1:** Identify the relevant electricity systems;
- Step 2:** Choose whether to include off-grid power plants in the project
- Step 3:** Select a method to determine the operating margin (OM);
- Step 4:** Calculate the operating margin emission factor according to the
- Step 5:** Calculate the build margin (BM) emission factor;
- Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identity the relevant electricity systems

Ministry of Ecology and Environment of the People’s Republic of China has published a delineation of the project electricity system and connected electricity systems, so the project adopts the delineation of project electricity system and connected electricity system published by Ministry of Ecology and Environment of the People’s Republic of China. The power generated by the project displaces the equivalent electricity generated by the East China Power Grid. The East China Power Grid is a larger regional grid, which consists of six sub-grids: Shanghai City, Jiangsu Province, Zhejiang Province, Anhui Province, Fujian Province.

In addition, there is net imported power to the East China Power Grid from the North China Power Grid, Northwest China Power Grid and Central China Power Grid. According to the “Tool to calculate the emission factor for an electricity system”, use one of the following

options to determine the CO₂ emission factor for net electricity imports from a connected electricity system:

(a) 0tCO₂/MWh; or

(b) The simple operating margin emission rate of the exporting grid, determined as described in Step 4 section 6.4.1, if the conditions for this method, as described in Step 3 below, apply to the exporting grid; or

(c) The simple adjusted operating margin emission rate of the exporting grid, determined as described in Step 4 section 6.4.2 below; or

(d) The weighted average operating margin (OM) emission rate of the exporting grid, determined as described in Step 4 section 6.4.4 below.

The PD will choose option (b) to calculate the CO₂ emission factor for net electricity imports from the East China Power Grid.

According to the available data of 2016-2018, the corresponding marginal emission factors of electricity are calculated, and weighted average is carried out to obtain the marginal emission factors of electricity.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional).

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Following the calculation of the Ministry of Ecology and Environment of the People's Republic of China, and statistical data is available, Option I is chosen.

Step 3. Select a method to determine the operating margin (OM)

“Tool to calculate the emission factor for an electricity system (Version 7.0)” offers four methods for the calculation of the operating margin emission factor(s) (EF_{grid,OM,y}):

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

Method (a) -Simple OM is chosen for calculation and low-cost/must-run resources constitute less than 50% of the total grid generation in average of the five most recent years¹².

For simple OM, the emission factor can be calculated using either of the two following data vintages:

- (a) Ex ante option: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the VCS-PD to the DOE for validation. For off-grid power plants, use a single calendar year within the 5 most recent calendar years prior to the time of submission of the VCS-PD for validation;
- (b) Ex post option: If the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required to calculate the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year y-1 may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year proceeding the previous year y-2 may be used. The same data vintage (y, y-1 or y-2) should be used throughout all crediting periods

Project participant employs ex ante option for its operation margin calculation.

Step 4. Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated by one of the following two options:

- (a) Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit; or

¹² 2019 Bulletin on the Baseline Emission Factors of the China Grids

(b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Option B can only be used if:

- (i) The necessary data for Option A is not available: and
- (ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and
- (iii) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2).

Since the data of each power plant/unit is unavailable, Option A is not applicable to the project. The project adopts Option B to calculate the operating margin emission factor ($EF_{grid,OM,y}$) of CCPG.

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_y} \quad \text{Equation (6)}$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO ₂ emission factor of fossil fuel type i in year y (tCO ₂ e/GJ), 2006 IPCC Guidelines for default values
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must run power plants/units in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

Regarding parameter selection, local values of $NCV_{i,y}$ and $EF_{CO2,i,y}$ should be used where available. If no such values are available, IPCC default values are preferable. The Net Calorific Value ($NCV_{i,y}$) of each type of fossil fuel used in the calculation comes from China

Energy Statistic Yearbook 2018. Emission factor ($EF_{CO_2,i,y}$) of each type of fossil fuel come from IPCC 2006 default values.

The operating margin emission factor for 2016, 2017 and 2018 are calculated based on the data above. The three-year average is calculated as a weighted average of the emission factors. The Operational Margin Emission Factor is $0.8587tCO_2/MWh$.

Step 5. Calculate the build margin emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

- (a) Option 1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of VCS PD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.
- (b) Option 2: For the first crediting period, the build margin emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The project applies option 1 to calculate the build margin emission factor ex-ante.

According to the “Tool to calculate the emission factor for an electricity system”, the sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

- (a) Identify the set of five power units, excluding power units registered as VCS project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) and determine their annual electricity generation ($AEG_{SET-5-units}$, in MWh);
- (b) Determine the annual electricity generation of the project electricity system, excluding power units registered as VCS project activities (AEG_{total} , in MWh). Identify the set of power

units, excluding power units registered as VCS project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET_{\geq 20\%}}$, in MWh);

(c) From $SET_{5-units}$ and $SET_{\geq 20\%}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. Ignore steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as VCS project activity, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set (SET_{sample}) the annual electricity generation ($AEG_{SET_{sample}}$, in MWh);

If the annual electricity generation of that set is comprised at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET_{sample}} \geq 0.2 \times AEG_{total}$), then use the sample group SET_{sample} to calculate the build margin. Ignore steps (e) and (f).

(e) Include in the sample group SET_{sample} the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample > 10yrs}$)

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad \text{Equation (7)}$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

According to the tools, the generation capacity of and the proportions of different generating technologies used in the mostly recent capacity additions would be calculated at first. The weight of each generating technology used in capacity additions can then be worked out and finally the Build Margin emission factor would be calculated with the commercial optimal efficiencies of the generating technologies.

The following method is adopted for the BM calculation. The first step is to use the most recently available data on the energy balance to calculate the proportion of CO₂ emissions from solid, liquid, and gas fuels used for power generation in total CO₂ emissions for the CCPG in 2017. Then this proportion is applied as a weight and combined with the emission factors of the best commercially available technologies to calculate the thermal emission factor of the CCPG. Finally, the BM factor of the CCPG can be determined by multiplying its thermal emission factor by the capacity-weighted share of thermal power in the newly added 20% of the installed capacity

(1) Calculate the proportion of CO₂ emissions from solid, liquid, and gas fuels used for power generation in total CO₂ emissions for the CCPG in the base year (2015)

$$\lambda_{Coal,y} = \frac{\sum_{i \in COAL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}} \quad \text{Equation (8)}$$

$$\lambda_{Oil,y} = \frac{\sum_{i \in OIL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}} \quad \text{Equation (9)}$$

$$\lambda_{Gas,y} = \frac{\sum_{i \in GAS,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}} \quad \text{Equation (10)}$$

Where:

$F_{i,j,y}$	=	The amount of fuel i (in a mass or volume unit) consumed by province
-------------	---	--

$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i (GJ/mass or volume unit)

$COAL, OIL \& GAS$ = The aggregation of various kinds of coal, oil, and gas as fossil fuels.

(2) Calculate the corresponding emission factor for fossil fuel fired power generation.

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal,Adv} + \lambda_{Oil} \times EF_{Oil,Adv} + \lambda_{Gas} \times EF_{Gas,Adv} \quad \text{Equation (11)}$$

Where:

$EF_{Coal,adv}$, $EF_{Oil,adv}$, $EF_{Gas,adv}$ are the emission factors of coal, oil and gas-fired power generation with efficiency levels of the best commercially available technology in China in the previous three years.

(3) Calculate the Building Margin emission factor

$$EF_{BM,y} = \frac{CAP_{Thermal}}{CAP_{Total}} \times EF_{Thermal} \quad \text{Equation (12)}$$

Where:

$CAP_{Thermal}$ = Total capacity additions while

CAP_{Total} = Capacity additions of thermal power

Following the four steps above, the build margin emission factor, $EF_{grid,BM,y}$ of the CCPG is calculated to be 0.2854 tCO₂/MWh.

Step 6. Calculate the combined margin emission factor

Combined Margin emission factor ($EF_{grid,CM,y}$) is calculated as the weighted average of the operating margin emission factor ($EF_{grid,OM,y}$) and the build margin emission factor ($EF_{grid,BM,y}$), where the weights ω_{OM} and ω_{BM} , by default, are 0.5 and 0.5 in the first crediting period, and $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ are calculated as described above and are expressed in tCO₂/MWh.

$$EF_{grid,CM,y} = \omega_{OM} \times EF_{grid,OM,y} + \omega_{BM} \times EF_{grid,BM,y} \quad \text{Equation (13)}$$

$$EF_{grid,CM,y} = 0.5 \times 0.8587 + 0.5 \times 0.2854 = 0.57205 \text{ (tCO}_2\text{/MWh)}$$

The $EF_{OM,y}$, $EF_{BM,y}$ and $EF_{grid,CM,y}$ are ex-ante calculation and are fixed during the credit period.

By combining all the above equations, the ex-ante estimate of the emission reduction is :

$$BE_y = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y}$$

Equation (14)

$$\begin{aligned} BE_y &= 313,608 \text{ m}^3 \times 0.056805 \text{ t/m}^3 \times 76.33\% \times 0.25 \text{ t CH}_4/\text{t} \\ &\quad \text{COD} \times 0.89 \times 28 + 11.847 \text{ MWh} \times 0.57205 \text{ tCO}_2\text{e/MWh} \\ &= 67,771 \text{ tCO}_2\text{e} + 6,777 \text{ tCO}_2\text{e} = 74,548 \text{ tCO}_2\text{e} \end{aligned}$$

4.2 Project Emissions

$$PE_y = PE_{BR,y} + PE_{EC,y}$$

Equation (15)

Where:

- PE_y = Project emissions in the year y (t CO₂e)
- $PE_{BR,y}$ = Project emissions from biogas recovery in the year y (t CO₂e).
- $PE_{EC,y}$ = Project emissions from electricity consumption in the year y (t CO₂e).

Project emissions from biogas recovery calculation:

$$PE_{BR,y} = PE_{power,y} + PE_{ww,treatment,y} + PE_{fugative,y} + PE_{ww,discharge,y} + PE_{s,treatment,y} + PE_{s,final,y} + PE_{fugative,y} + PE_{biomass,y} + PE_{flaring,y}$$

Equation (16)

Where:

- $PE_{BR,y}$ = Project emissions from biogas recovery in the year y (t CO₂e)
- $PE_{power,y}$ = Emissions from electricity or fuel consumption in the year y (t CO₂e).
- $PE_{ww,treatment,y}$ = Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO₂e).
- $PE_{ww,discharge}$ = Methane emissions from degradable organic carbon in treated wastewater in year y (tCO₂e).
- $PE_{fugative,y}$ = Methane emissions from biogas release in capture systems in year y , calculated as per paragraph 40 (t CO₂e).
- $PE_{s,treatment,y}$ = Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO₂e).
- $PE_{s,final,y}$ = Methane emissions from anaerobic decay of the final sludge produced in year y (t CO₂e).
- $PE_{fugative,y}$ = Methane emissions from biogas release in capture systems in year y (t CO₂e).
- $PE_{biomass,y}$ = Methane emissions from biogas stored under anaerobic conditions.
- $PE_{flaring,y}$ = Methane emissions due to incomplete flaring in year y (t CO₂e).

There is no flare used to destroy the leachate biogas, therefore, $PE_{flaring,y}$ is equal to 0.

The sludge is controlled combusted by sent to incinerator. Project activity does not involve sludge treatment, therefore, $PE_{s,treatment,y}$ and $PE_{s,final,y}$ is equal to 0.

These emissions shall be calculated as per paragraph 26 of the methodology, If recovered biogas in the baseline is used to power auxiliary equipment it should be taken into account accordingly, using zero as its emission factor. The electricity used by project activity will be firstly supplied by the gas engines which use biogas as fuel, to import electricity from the grid is the second option when the gas engines does not operate. The project deliver the electricity to the project activity before delivery to the grid. Then the chance that the project activity will use electricity from grid is very low. Therefore, $PE_{power,y}$ can be considered as zero.

Only when the biogas plant need to import the electricity from grid the $PE_{power,y}$ will be taken into account , which can be calculated from the same equation (4) of the baseline calculation formula for AMS I.D.

$$PE_{power,y} = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (17)}$$

The water treatment system remains unchanged, and only methane capture system was installed. Therefore, $PE_{ww,treatment,y}$, $PE_{biomass,y}$ and $PE_{ww,discharge,y}$ is equal to 0.

Therefore, the baseline emissions equation is simplified to:

$$PE_{BR,y} = PE_{fugitive,y} \quad \text{Equation (18)}$$

$$PE_{fugitive,ww,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH_4} \quad \text{Equation (19)}$$

Where:

CFE_{ww} = Capture efficiency of the biogas recovery equipment in the wastewater treatment systems (a default value of 0.9 shall be used)

$MEP_{ww,treatment,y}$ = Methane emission potential of wastewater treatment systems equipped with biogas recovery system in year y (t)

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} \times UF_{PJ} \times \sum_k COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k} \quad \text{Equation (20)}$$

Where:

$COD_{removed,PJ,k,y}$	=	The chemical oxygen demand removed ¹³ by the treatment system k of the project activity equipped with biogas recovery in the year y (t/m ³)
$MCF_{ww,treatment,PJ,k}$	=	Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment (MCF values as per Table 2 above)
UF_{PJ}	=	Model correction factor to account for model uncertainties (1.12)

The Methane Correction Factor (MCF) shall be determined based on the table 4.1. IPCC default values 7 for Methane Correction Factor (MCF).

$$MCF_{ww,treatment,PJ,k} = 0.8$$

$$\begin{aligned} MEP_{ww,treatment,y} &= 313,608 * 0.041880 * 0.25 * 0.8 * 1.12 \\ &= 2,942 \text{ tCH}_4/\text{yr} \end{aligned}$$

$$\begin{aligned} PE_{BR,y} &= (1 - 0.9) * 2942 * 28 \\ &= 8,238 \text{ tCO}_2\text{e}/\text{yr} \end{aligned}$$

Project emissions from electricity consumption calculation:

$PE_{EC,y}$ is determined by “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (Version 3.0) as follow:

$$PE_{EC,y} = \sum EC_{PJ,j,y} * EF_{EF,j,y} * (1 + TDL_{j,y}) \quad \text{Equation(21)}$$

Where:

$PE_{EC,y}$	=	Project emissions from electricity consumption in year y (tCO ₂ /yr)
$EC_{PJ,j,y}$	=	Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)
$EF_{EF,j,y}$	=	Emission factor for electricity generation for source j in year y (tCO ₂ /MWh)
$TDL_{j,y}$	=	Average technical transmission and distribution losses for providing electricity to source j in year y

¹³ Difference between the inflow COD and the outflow COD.

The “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” provides 3 scenarios for different sources of electricity consumption. Since the electricity generated through the project is consumed solely from grid (CCPG), scenario A is applicable. In the case of applying scenario A, two options are available for determining emission factor. For this project, Option A1 ($EF_{EF,j,y} = EF_{grid,CM,y}$) is chosen.

For simplification, $PE_{EC,y}$ will be 0 in the ex-ante calculation.

4.3 Leakage

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.H. and AMS-I.D.

4.4 Net GHG Emission Reductions and Removals

According to AMS-III.H., the emission reduction achieved by the project activity can be estimated ex-ante in the PD by:

$$ER_{y,estimated} = BE_y - PE_y - LE_y \quad \text{Equation (22)}$$

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$ER_{y,estimated} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad \text{Equation (23)}$$

The data used for calculation is from Environmental Impact Assessment and project operation daily.

$$\begin{aligned} ER_{y,estimated} &= 313608 \times 0.056805 \times 76.33\% \times 0.8 \times 0.25 \times 0.89 \times 28 + 11,847 \times 0.57205 - 8,238 - 0 \\ &= 67,771 + 6,777 - 8,238 - 0 \\ &= 66,310 \text{ tCO}_2/\text{yr} \end{aligned}$$

Emission reductions (ex-post)

According to AMS-III.H., The actual emission reduction achieved by the project activity during the crediting period will be calculated as:

$$ER_{y,ex,post} = \min((BE_{y,ex,post} - PE_{y,ex,post} - LE_{y,ex,post})(MD_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex,post}))$$

Equation (24)

Where:

- $ER_{y,ex,post}$ = Emission reductions achieved by the project activity based on monitored values for year y (t CO₂e)
- $BE_{y,ex,post}$ = Baseline emissions calculated as per paragraph 25 using ex post monitored values
- $PE_{y,ex,post}$ = Project emissions calculated as per paragraph 39 using ex post monitored values
- MD_y = Methane captured and destroyed/gainfully used by the project activity in the year y (t CO₂e)

In the case of flaring/combustion MD_y will be measured using the conditions of the flaring process:

$$MD_y = BG_{burnt,y} \times W_{CH_4,y} \times D_{CH_4} \times FE \times GWP_{CH_4}$$

Equation (25)

Where:

- $BG_{burnt,y}$ = Biogas¹⁴ combusted in year y (m³)
- $W_{CH_4,y}$ = Methane content¹³ of the biogas in the year y (volume fraction)
- D_{CH_4} = Density of methane at the temperature and pressure of the biogas in the year y (t/m³)
- FE = Flare efficiency in year y (fraction). If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
21/01/2022-31/12/2022	70,463	7,786	0	62,677
2023	74,548	8,238	0	66,310

¹⁴ Biogas volume and methane content measurements shall be on the same basis (wet or dry).

2024	74,548	8,238	0	66,310
2025	74,548	8,238	0	66,310
2026	74,548	8,238	0	66,310
2027	74,548	8,238	0	66,310
2028	74,548	8,238	0	66,310
2029	74,548	8,238	0	66,310
2030	74,548	8,238	0	66,310
2031	74,548	8,238	0	66,310
01/01/2032- 20/01/2032	4,084	451	0	3,633
Total	745,479	82,379	0	663,100

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$MCF_{ww,treatment,BL,i}$
Data unit	-
Description	Methane correction factor for baseline wastewater treatment systems i
Source of data	MCF values as per Table 2 of AMS-III.H. V.19
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	Default values from chapter 6 of volume 5. Waste in 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\eta_{\text{COD, BL, i}}$
Data unit	%
Description	COD removal efficiency of the baseline treatment system
Source of data	Operation report
Value applied	76.33%
Justification of choice of data or description of measurement methods and procedures applied	The COD removal efficiency is based on COD values for baseline system before implementation of project activity. Detailed data is shown in APPENDIX Table 5.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$B_{0, \text{ww}}$
Data unit	kg CH ₄ /kg COD
Description	Methane producing capacity of the wastewater
Source of data	Paragraph 29 of AMS-III.H V.19
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value applies in line with Equation 2 of AMS-III.H V.19
Purpose of Data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF_{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	Paragraph 29 of AMS-III.H V.19

Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	As specified in Paragraph 29 of AMS-III.H V.19
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF_{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H V.19
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	As specified in methodology AMS-III.H V.19
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$MCF_{ww,treatment,PJ,k}$
Data unit	-
Description	Methane correction factor for project wastewater treatment system k
Source of data	Table 2 of AMS-III.H V.19
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	As specified in methodology AMS-III.H V.19

Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC
Value applied	Default value of 28. Shall be updated according to any future COP/CMP decisions
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	CFE _{ww}
Data unit	-
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	Default value as per paragraph 40 Eq 10 of AMS-III.H V.19
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,OM,y}
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Data unit	tCO ₂ /MWh
Description	Operation margin emission factor of CCPG
Source of data	2019 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.8587
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Electric Yearbook (2016-2018), Electricity Industry Statistics (2015-2017), China Energy Statistic Yearbook (2016-2018) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,BM,y}
Data unit	tCO ₂ /MWh
Description	Build margin emission factor of CCPG
Source of data	2019 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.2854
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Electric Yearbook (2016-2018), Electricity Industry Statistics (2015-2017), China Energy Statistic Yearbook (2016-2018) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	TDL _{j,y}
Data unit	%
Description	Average technical transmission and distribution losses for providing electricity to source j,k in year y

Source of data	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)
Value applied	20%
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	D_{CH_4}
Data unit	tCH ₄ /m ³
Description	Methane density
Source of data	-
Value applied	0.0007168
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	At standard temperature and pressure (0 degree Celsius and 1,013 bar) the density of methane is 0.0007168 tCH ₄ /m ³ CH ₄

Data / Parameter	FE
Data unit	%
Description	Flare efficiency in year y
Source of data	Default value as per paragraph 45 Eq 10 of AMS-III.H V.19
Value applied	100
Justification of choice of data or description of	If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied

measurement methods and procedures applied	
Purpose of Data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$Q_{ww,i,y}$										
Data unit	m ³ /month										
Description	The flow of wastewater										
Source of data	Continuous flow meter										
Description of measurement methods and procedures to be applied	Measured continuously by flow meters installed before entering the treatment system. The flow of wastewater is the sum of F1 and F2.										
Frequency of monitoring/recording	Parameter monitored continuously but aggregated monthly and annually for calculations.										
Value applied	313,608										
Monitoring equipment	Continuous Flow Meter <table border="1"> <thead> <tr> <th></th><th>Accuracy</th><th>Serial Number</th></tr> </thead> <tbody> <tr> <td>North zone Flow Meter F1</td><td>0.5</td><td>R70DB318000</td></tr> <tr> <td>South zone Flow Meter F2</td><td>0.5</td><td>S9040018000</td></tr> </tbody> </table>			Accuracy	Serial Number	North zone Flow Meter F1	0.5	R70DB318000	South zone Flow Meter F2	0.5	S9040018000
	Accuracy	Serial Number									
North zone Flow Meter F1	0.5	R70DB318000									
South zone Flow Meter F2	0.5	S9040018000									
QA/QC procedures to be applied	Zhengzhou Dongxing Environmental Protection Energy Co., Ltd. employs an internal QA audit process that ensures monitoring activities are conducted in accordance with the monitoring plan and verifies the accuracy of data reported. The flow meter will be operated and calibrated according to manufacturer's specifications but no less than every 3 years). A calibration/service log will be maintained for the flow meter.										

Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	BG _{burnt,y}							
Data unit	Nm ³							
Description	Biogas volume in year y							
Source of data	Daily records of the flow meter F3							
Description of measurement methods and procedures to be applied	Measured continuously by flow meters installed before entering the treatment system.							
Frequency of monitoring/recording	Measured continuously by a flow meter, and recorded at least once per hour. Data to be aggregated monthly and yearly.							
Value applied	553.6 (based on EIA and all biogas is combusted in gas engines)							
Monitoring equipment	Continuous Flow Meter <table border="1"> <thead> <tr> <th></th><th>Accuracy</th><th>Serial Number</th></tr> </thead> <tbody> <tr> <td>Flow Meter F3</td><td>1.5</td><td>S91A7C02000</td></tr> </tbody> </table>			Accuracy	Serial Number	Flow Meter F3	1.5	S91A7C02000
	Accuracy	Serial Number						
Flow Meter F3	1.5	S91A7C02000						
QA/QC procedures to be applied	The calibration is being done once a year.							
Purpose of data	Calculation of baseline and project emissions							
Calculation method	-							
Comments	Flow meters automatically measure temperature and pressure, expressing LFG volumes in Normalized cubic meters (Nm ³). Data are archived during the crediting period and kept until two years after.							

Data / Parameter	W _{CH4,y}			
Data unit	%			
Description	Methane content in biogas in the year y			
Source of data	Daily records of the gas analyzer meter			
Description of measurement methods and procedures to be applied	Measured directly by continuous gas analyzer.			
Frequency of monitoring/recording	Measurements are recorded electronically at least once per day. The data are measured and archived electronically.			
Value applied	65% (assumed based on feasibility study report)			
Monitoring equipment	Gas analyzer			
	Name	Type	Accuracy class	Serial number
	Gas analyzer	BIOGAS 5000	0.5%	G502590
QA/QC procedures to be applied	Gas analyzer is calibrated once a year.			
Purpose of data	Calculation of baseline and project emissions			
Calculation method	-			
Comments	-			

Data / Parameter	$COD_{ww,untreated,y}$
Data unit	t COD/m ³
Description	The chemical oxygen demand of the wastewater before the treatment system affected by the project activity
Source of data	Sampling analysis
Description of measurement methods	The COD content will be analyzed using a spectrophotometric method. The results will be logged in the plant operation report on a daily basis.

and procedures to be applied	
Frequency of monitoring/recording	Measured and recorded every day. The confidence/precision level of 90/10 shall be attained.
Value applied	The data below are calculated ex-ante. Detailed data is shown in APPENDIX Table 5.
Monitoring equipment	UV756 UV spectrophotometer
QA/QC procedures to be applied	Measure the COD in accordance with Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017).
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	COD _{ww,treated,y}
Data unit	t COD/m ³
Description	The chemical oxygen demand of the wastewater after the treatment system affected by the project activity
Source of data	Sampling analysis
Description of measurement methods and procedures to be applied	The COD content will be analyzed using a spectrophotometric method. The results will be logged in the plant operation report on a daily basis.
Frequency of monitoring/recording	Measured and recorded every day. The confidence/precision level of 90/10 shall be attained.
Value applied	The data below are calculated ex-ante. Detailed data is shown in APPENDIX Table 5.

Monitoring equipment	UV756 UV spectrophotometer
QA/QC procedures to be applied	Measure the COD in accordance with Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017).
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$EG_{PJ, facility, y}$												
Data unit	MWh												
Description	Quantity of net electricity generated supplied by the project plant/unit in year y												
Source of data	Electricity meter E												
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter (bi-directional) installed at the project site. All data will be monitored and archived electronically. Double check by receipt of electricity sales.												
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.												
Value applied	The data below are calculated ex-ante. <table border="1"> <thead> <tr> <th>Year</th><th>Quantity of net electricity generated supplied by the project plant (MWh)</th></tr> </thead> <tbody> <tr> <td>21/01/2022-31/12/2022</td><td>11,198</td></tr> <tr> <td>2023</td><td>11,847</td></tr> <tr> <td>2024</td><td>11,847</td></tr> <tr> <td>2025</td><td>11,847</td></tr> <tr> <td>2026</td><td>11,847</td></tr> </tbody> </table>	Year	Quantity of net electricity generated supplied by the project plant (MWh)	21/01/2022-31/12/2022	11,198	2023	11,847	2024	11,847	2025	11,847	2026	11,847
Year	Quantity of net electricity generated supplied by the project plant (MWh)												
21/01/2022-31/12/2022	11,198												
2023	11,847												
2024	11,847												
2025	11,847												
2026	11,847												

	2027	11,847
	2028	11,847
	2029	11,847
	2030	11,847
	2031	11,847
	01/01/2032-20/01/2032	649
Monitoring equipment	Electricity meter E shown in the monitoring system as below.	
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.	
Purpose of data	Calculation of baseline emissions	
Calculation method	-	
Comments	-	

Data / Parameter	$EC_{PJ,y}$
Data unit	MWh
Description	Amount of electricity consumed by the project activity in year y
Source of data	Electricity meter E
Description of measurement methods and procedures to be applied	Sources of consumption include electricity consumed for the operation of the biogas capture system, for any processing and upgrading of the biogas, for transportation of the biogas to the flare or power generators, etc.
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.
Value applied	For simplification, the data used in ex ante calculation is 0.
Monitoring equipment	Electricity meter E shown in the monitoring system as below.
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.

Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the biogas
Source of data	-
Description of measurement methods and procedures to be applied	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas.
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	-
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	°C
Description	Temperature of the biogas
Source of data	-
Description of measurement methods and procedures to be applied	. The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs

	the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	-
Calculation method	-
Comments	-

5.3 Monitoring Plan

1. The requirement of monitoring plan

The project participants will monitor the emission reductions by methods, indicators, and frequency required by Monitoring Methodology AMS-III.H. (Version 19.0) and AMS-I.D (Version 18.0) to ensure project ERs are measurable and real. The monitoring methodology is based on direct measurement of the amount of leachate biogas captured and destroyed by the project and electricity generating units.

2. Responsibilities of operational and management structure

The project participant will implement this monitoring plan. The plan could be revised according to suggestions from VVB and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

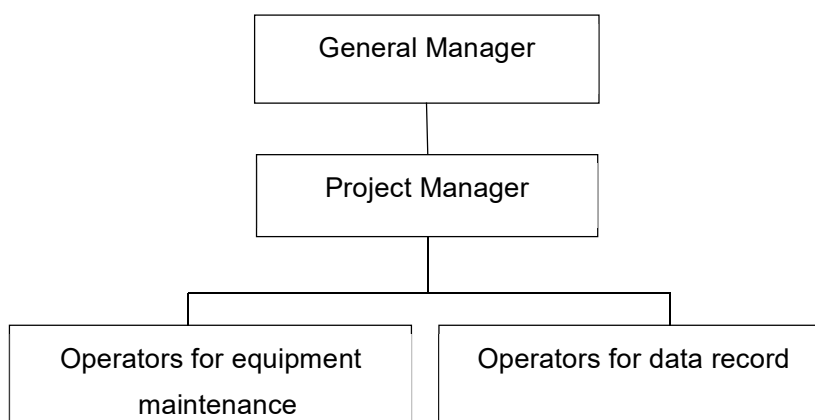


Figure 5.1 Operational and management structure

(1) Principal of the monitoring procedure

The general manager of the project is the leader of the monitoring tasks: he sets out the responsibility of everyone in the monitoring system, and establishes the related documents. The general manager ensures that staff in the monitoring system has the ability to deal with the assigned tasks.

(2) Executive person of the monitoring procedure

Project Manager: a Project Manager is appointed, specifically responsible for training, checking the daily operation, reporting forms and archiving emergency situation reports. The VCS Manager reports monthly to the General Manager (GM) about the project performance and monitored data. In the event that non-conformance in the performance to the mentioned procedures and/or functioning problems of the monitoring equipment are identified, the VCS manager will inform the GM about the situation and work out relevant measures to be taken. The VCS manager will also be responsible for aggregating the monitored data monthly and yearly, archiving and keeping data during the crediting period and two years after.

(3) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day. They will be in charge of data supervision, filling operation report forms and, checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system

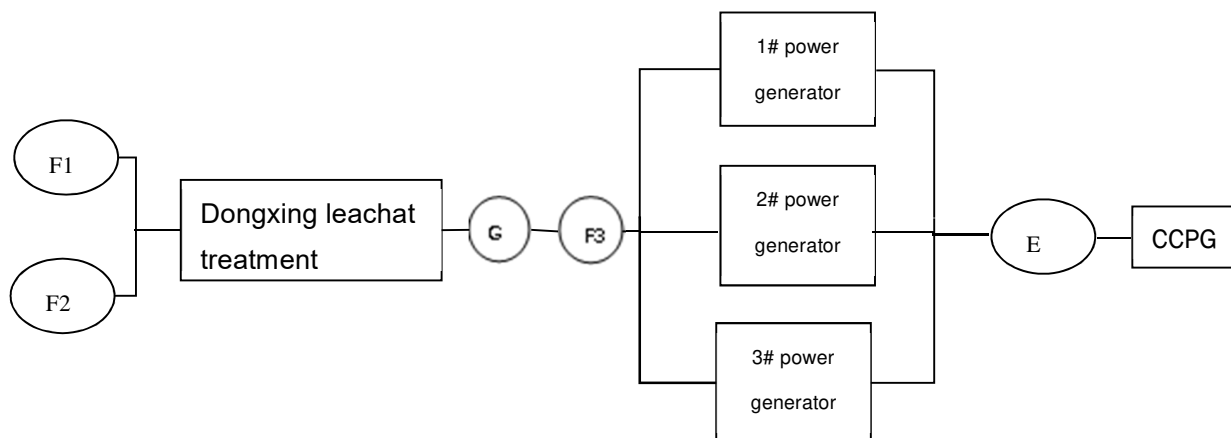


Figure 5.2 Monitoring system

Main parameters:

E	Electricity meter (main meter, bi-directional) to continuously measure the electricity supplied to the CCPG and the electricity supplied by the grid
F1	Flow meter to continuously measure the flow of north zone wastewater before entering the treatment system.
F2	Flow meter to continuously measure the flow of south zone wastewater before entering the treatment system.
F3	Flow meter to continuously measure flow rate of total biogas, with automatic measure adjustment for temperature and pressure.
G	Gas analyzer to continuously measure methane fraction in biogas (W_{CH_4}).

Net electricity exported to the grid:

Electricity meter E is the main meter for monitoring the net electricity supplied to and imported from the grid. Therefore, the net electricity supplied to the grid is calculated as: $EG_{OUT} - EG_{IN}$.

The meter E is installed at the project site and sends data to the DCS. The data was checked daily and totalized monthly. Electricity receipts and invoices was used as cross check during verification. E was calibrated by a qualified third party according to the industry standards and applicable regulations.

4. QA/QC

In order to ensure monitoring plan with high quality, QA/QC measures are carried out in monitoring data recording and checking, equipment calibrating and staff training.

Data recording: all data collected are recorded in electronic files which are regularly backed

up. The data are checked by the specific staff every day and reported to General Manager by Project manager monthly.

Equipment calibration and maintenance: Project equipment are subject to regular maintenance and testing according to technical specifications from the manufactures to ensure accuracy and good performance.

If there is a problem with the project's monitoring equipment, the relevant data will be discarded during this period.

5. Emergency procedures

Once a meter in fault, it shall be immediately repaired or replaced with another calibrated meter by a professional engineer, and the biogas or electricity generated during the period of erroneous measurement and replacement of the fault meter shall not be accounted for conservative consideration.

APPENDIX

Table 5: Additional information for ex-ante estimation

Date	COD _{ww,untreated,south} (mg/L)	COD _{ww,untreated,north} (mg/L)	COD _{ww,untreated,y} (mg/L)	COD _{ww,treated,south} (mg/L)	COD _{ww,treated,north} (mg/L)	COD _{ww,treated,y} (mg/L)	COD _{ww,untreated,y} (t/m ³)	COD _{ww,treated,y} (t/m ³)	η _{COD, BL,i} (%)
01-Jan-21	/	/	/	/	/	/	/	/	/
02-Jan-21	/	/	/	/	/	/	/	/	/
03-Jan-21	64,200	62,300	63,250	15,356	15,735	15,546	0.063250	0.015546	75.42 %
04-Jan-21	/	/	/	/	/	/	/	/	/
05-Jan-21	/	/	/	/	/	/	/	/	/
06-Jan-21	59,100	62,900	61,000	14,541	14,273	14,407	0.061000	0.014407	76.38 %
07-Jan-21	/	/	/	/	/	/	/	/	/
08-Jan-21	/	/	/	/	/	/	/	/	/
09-Jan-21	/	/	/	/	/	/	/	/	/
10-Jan-21	57,300	61,000	59,150	15,992	13,965	14,979	0.059150	0.014979	74.68 %
11-Jan-21	/	/	/	/	/	/	/	/	/
12-Jan-21	/	/	/	/	/	/	/	/	/
13-Jan-21	/	65,900	65,900	/	14,448	14,448	0.065900	0.014448	78.08 %
14-Jan-21	/	/	/	/	/	/	/	/	/
15-Jan-21	/	/	/	/	/	/	/	/	/
16-Jan-21	/	/	/	/	/	/	/	/	/

17-Jan-21	63,350	66,061	64,706	14,955	15,213	15,084	0.064706	0.015084	76.69 %
18-Jan-21	62,963	67,223	65,093	14,395	14,792	14,594	0.065093	0.014594	77.58 %
19-Jan-21	67,611	63,033	65,322	15,157	15,870	15,514	0.065322	0.015514	76.25 %
20-Jan-21	65,287	55,674	60,481	14,952	15,249	15,101	0.060481	0.015101	75.03 %
21-Jan-21	65,674	58,385	62,030	14,428	15,067	14,748	0.062030	0.014748	76.23 %
22-Jan-21	66,061	57,611	61,836	14,803	14,121	14,462	0.061836	0.014462	76.61 %
23-Jan-21	62,189	56,449	59,319	14,216	15,456	14,836	0.059319	0.014836	74.99 %
24-Jan-21	60,639	58,385	59,512	15,148	15,365	15,257	0.059512	0.015257	74.36 %
25-Jan-21	62,963	56,449	59,706	14,031	15,668	14,850	0.059706	0.014850	75.13 %
26-Jan-21	64,125	58,316	61,221	14,444	14,398	14,421	0.061221	0.014421	76.44 %
27-Jan-21	62,963	61,801	62,382	14,423	14,672	14,548	0.062382	0.014548	76.68 %
28-Jan-21	65,674	61,027	63,351	15,598	14,018	14,808	0.063351	0.014808	76.63 %
29-Jan-21	62,189	64,125	63,157	13,884	14,396	14,140	0.063157	0.014140	77.61 %
30-Jan-21	59,090	55,217	57,154	14,148	15,128	14,638	0.057154	0.014638	74.39 %
31-Jan-21	54,830	52,963	53,897	14,891	15,671	15,281	0.053897	0.015281	71.65 %
01-Feb-21	57,611	54,125	55,868	14,568	14,859	14,714	0.055868	0.014714	73.66 %
02-Feb-21	55,287	53,738	54,513	15,197	15,848	15,523	0.054513	0.015523	71.52 %
03-Feb-21	52,963	51,801	52,382	15,671	15,113	15,392	0.052382	0.015392	70.62 %
04-Feb-21	52,963	51,801	52,382	14,230	15,598	14,914	0.052382	0.014914	71.53 %
05-Feb-21	52,841	51,770	52,306	15,845	14,574	15,210	0.052306	0.015210	70.92 %
06-Feb-21	55,674	57,998	56,836	14,385	15,130	14,758	0.056836	0.014758	74.03 %
07-Feb-21	56,836	52,189	54,513	15,597	15,812	15,705	0.054513	0.015705	71.19 %

08-Feb-21	59,547	59,934	59,741	15,695	15,156	15,426	0.059741	0.015426	74.18 %
09-Feb-21	59,934	54,900	57,417	15,892	14,143	15,018	0.057417	0.015018	73.84 %
10-Feb-21	62,645	61,096	61,871	15,964	15,521	15,743	0.061871	0.015743	74.56 %
11-Feb-21	62,258	61,484	61,871	13,960	15,976	14,968	0.061871	0.014968	75.81 %
12-Feb-21	62,645	60,709	61,677	14,165	14,826	14,496	0.061677	0.014496	76.50 %
13-Feb-21	60,709	61,096	60,903	14,685	14,545	14,615	0.060903	0.014615	76.00 %
14-Feb-21	62,645	66,131	64,388	14,389	14,924	14,657	0.064388	0.014657	77.24 %
15-Feb-21	65,674	57,611	61,643	13,878	15,604	14,741	0.061643	0.014741	76.09 %
16-Feb-21	57,611	59,547	58,579	14,819	15,933	15,376	0.058579	0.015376	73.75 %
17-Feb-21	54,512	59,160	56,836	14,612	14,323	14,468	0.056836	0.014468	74.55 %
18-Feb-21	55,217	58,385	56,801	15,438	15,128	15,283	0.056801	0.015283	73.09 %
19-Feb-21	44,512	56,449	50,481	14,088	15,292	14,690	0.050481	0.014690	70.90 %
20-Feb-21	50,322	62,258	56,290	14,116	14,254	14,185	0.056290	0.014185	74.80 %
21-Feb-21	51,096	61,484	56,290	14,705	15,483	15,094	0.056290	0.015094	73.19 %
22-Feb-21	63,420	56,518	59,969	15,301	13,942	14,622	0.059969	0.014622	75.62 %
23-Feb-21	62,576	54,512	58,544	14,440	14,964	14,702	0.058544	0.014702	74.89 %
24-Feb-21	63,033	57,611	60,322	15,664	15,026	15,345	0.060322	0.015345	74.56 %
25-Feb-21	64,582	57,998	61,290	13,802	14,879	14,341	0.061290	0.014341	76.60 %
26-Feb-21	59,160	52,576	55,868	14,387	15,945	15,166	0.055868	0.015166	72.85 %
27-Feb-21	58,773	52,189	55,481	15,675	15,704	15,690	0.055481	0.015690	71.72 %
28-Feb-21	52,963	61,096	57,030	14,136	14,012	14,074	0.057030	0.014074	75.32 %
01-Mar-21	53,807	62,258	58,033	15,869	15,833	15,851	0.058033	0.015851	72.69 %

02-Mar-21	59,934	56,836	58,385	15,281	14,130	14,706	0.058385	0.014706	74.81 %
03-Mar-21	59,160	57,611	58,386	13,842	14,206	14,024	0.058386	0.014024	75.98 %
04-Mar-21	70,322	59,160	64,741	13,801	14,707	14,254	0.064741	0.014254	77.98 %
05-Mar-21	64,195	53,420	58,808	14,731	15,814	15,273	0.058808	0.015273	74.03 %
06-Mar-21	66,906	55,356	61,131	15,773	15,107	15,440	0.061131	0.015440	74.74 %
07-Mar-21	45,674	54,795	50,235	14,724	15,171	14,948	0.050235	0.014948	70.24 %
08-Mar-21	46,836	53,807	50,322	15,356	15,315	15,336	0.050322	0.015336	69.52 %
09-Mar-21	40,322	53,033	46,678	14,077	15,096	14,587	0.046678	0.014587	68.75 %
10-Mar-21	41,801	42,576	42,189	14,428	14,710	14,569	0.042189	0.014569	65.47 %
11-Mar-21	58,703	40,322	49,513	13,926	15,039	14,483	0.049513	0.014483	70.75 %
12-Mar-21	58,316	42,258	50,287	15,648	14,267	14,958	0.050287	0.014958	70.26 %
13-Mar-21	54,056	66,518	60,287	15,167	15,982	15,575	0.060287	0.015575	74.17 %
14-Mar-21	50,570	62,258	56,414	13,944	15,425	14,685	0.056414	0.014685	73.97 %
15-Mar-21	50,183	62,189	56,186	15,894	15,729	15,812	0.056186	0.015812	71.86 %
16-Mar-21	69,160	63,033	66,097	15,204	15,578	15,391	0.066097	0.015391	76.71 %
17-Mar-21	/	/	/	/	/	/	/	/	/
18-Mar-21	/	/	/	/	/	/	/	/	/
19-Mar-21	54,830	46,449	50,640	14,617	14,042	14,330	0.050640	0.014330	71.70 %
20-Mar-21	52,894	44,512	48,703	15,794	15,811	15,803	0.048703	0.015803	67.55 %
21-Mar-21	63,350	46,518	54,934	15,204	15,939	15,572	0.054934	0.015572	71.65 %
22-Mar-21	60,639	43,350	51,995	15,604	15,564	15,584	0.051995	0.015584	70.03 %
23-Mar-21	57,541	40,639	49,090	15,651	14,629	15,140	0.049090	0.015140	69.16 %

24-Mar-21	59,160	55,992	57,576	15,291	14,846	15,069	0.057576	0.015069	73.83 %
25-Mar-21	58,773	53,668	56,221	14,311	15,771	15,041	0.056221	0.015041	73.25 %
26-Mar-21	62,258	49,795	56,027	15,697	14,896	15,297	0.056027	0.015297	72.70 %
27-Mar-21	66,449	43,211	54,830	14,367	15,459	14,913	0.054830	0.014913	72.80 %
28-Mar-21	61,801	41,662	51,732	15,247	14,289	14,768	0.051732	0.014768	71.45 %
29-Mar-21	64,582	45,922	55,252	15,014	15,735	15,375	0.055252	0.015375	72.17 %
30-Mar-21	66,906	45,922	56,414	14,259	14,173	14,216	0.056414	0.014216	74.80 %
31-Mar-21	69,934	42,824	56,379	15,845	15,560	15,703	0.056379	0.015703	72.15 %
01-Apr-21	52,645	46,310	49,478	14,382	15,037	14,710	0.049478	0.014710	70.27 %
02-Apr-21	60,709	60,252	60,481	13,900	13,979	13,940	0.060481	0.013940	76.95 %
03-Apr-21	61,096	42,189	51,643	14,915	15,623	15,269	0.051643	0.015269	70.43 %
04-Apr-21	60,322	49,865	55,094	15,388	14,588	14,988	0.055094	0.014988	72.80 %
05-Apr-21	51,484	46,836	49,160	15,213	15,726	15,470	0.049160	0.015470	68.53 %
06-Apr-21	51,484	48,385	49,934	14,849	15,215	15,032	0.049934	0.015032	69.90 %
07-Apr-21	65,744	44,125	54,934	13,993	13,832	13,913	0.054934	0.013913	74.67 %
08-Apr-21	60,709	49,160	54,935	15,744	14,817	15,281	0.054935	0.015281	72.18 %
09-Apr-21	65,356	42,576	53,966	14,744	15,746	15,245	0.053966	0.015245	71.75 %
10-Apr-21	64,195	40,322	52,259	14,922	15,418	15,170	0.052259	0.015170	70.97 %
11-Apr-21	68,773	46,836	57,805	15,018	14,484	14,751	0.057805	0.014751	74.48 %
12-Apr-21	62,576	55,217	58,897	15,528	14,017	14,773	0.058897	0.014773	74.92 %
13-Apr-21	58,385	45,922	52,154	15,096	14,463	14,780	0.052154	0.014780	71.66 %
14-Apr-21	52,963	49,408	51,186	14,721	15,705	15,213	0.051186	0.015213	70.28 %

15-Apr-21	54,512	48,633	51,573	14,572	15,695	15,134	0.051573	0.015134	70.66 %
16-Apr-21	52,645	71,871	62,258	14,782	14,407	14,595	0.062258	0.014595	76.56 %
17-Apr-21	54,900	57,928	56,414	15,098	15,120	15,109	0.056414	0.015109	73.22 %
18-Apr-21	52,576	59,865	56,221	14,483	15,064	14,774	0.056221	0.014774	73.72 %
19-Apr-21	52,963	57,928	55,446	14,372	14,936	14,654	0.055446	0.014654	73.57 %
20-Apr-21	51,027	54,056	52,542	15,762	14,172	14,967	0.052542	0.014967	71.51 %
21-Apr-21	57,541	52,506	55,024	13,819	13,843	13,831	0.055024	0.013831	74.86 %
22-Apr-21	61,801	52,894	57,347	14,764	13,981	14,373	0.057347	0.014373	74.94 %
23-Apr-21	48,067	50,570	49,319	15,127	13,982	14,555	0.049319	0.014555	70.49 %
24-Apr-21	47,998	42,824	45,411	13,952	15,751	14,852	0.045411	0.014852	67.30 %
25-Apr-21	53,807	54,443	54,125	15,191	15,898	15,545	0.054125	0.015545	71.28 %
26-Apr-21	54,195	60,709	57,452	13,904	15,255	14,580	0.057452	0.014580	74.62 %
27-Apr-21	57,223	59,865	58,544	14,298	14,062	14,180	0.058544	0.014180	75.78 %
28-Apr-21	60,252	56,767	58,510	15,938	14,113	15,026	0.058510	0.015026	74.32 %
29-Apr-21	61,027	55,217	58,122	15,444	15,005	15,225	0.058122	0.015225	73.81 %
30-Apr-21	63,350	58,703	61,027	14,450	15,447	14,949	0.061027	0.014949	75.50 %
01-May-21	64,512	62,963	63,738	15,428	15,180	15,304	0.063738	0.015304	75.99 %
02-May-21	64,125	60,252	62,189	14,355	14,327	14,341	0.062189	0.014341	76.94 %
03-May-21	62,963	57,928	60,446	15,467	15,525	15,496	0.060446	0.015496	74.36 %
04-May-21	62,189	59,478	60,833	15,592	14,944	15,268	0.060833	0.015268	74.90 %

05-May-21	53,668	57,541	55,605	15,140	15,310	15,225	0.055605	0.015225	72.62 %
06-May-21	52,506	53,668	53,087	13,951	14,361	14,156	0.053087	0.014156	73.33 %
07-May-21	51,732	52,506	52,119	14,392	14,472	14,432	0.052119	0.014432	72.31 %
08-May-21	60,252	61,027	60,639	15,102	15,133	15,118	0.060639	0.015118	75.07 %
09-May-21	61,801	61,027	61,414	14,185	13,928	14,057	0.061414	0.014057	77.11 %
10-May-21	67,611	60,639	64,125	14,216	15,181	14,699	0.064125	0.014699	77.08 %
11-May-21	59,478	58,316	58,897	14,592	14,872	14,732	0.058897	0.014732	74.99 %
12-May-21	60,252	58,703	59,478	14,419	15,030	14,725	0.059478	0.014725	75.24 %
13-May-21	55,605	51,732	53,669	14,774	15,850	15,312	0.053669	0.015312	71.47 %
14-May-21	52,506	51,344	51,925	15,239	15,185	15,212	0.051925	0.015212	70.70 %
15-May-21	54,443	51,732	53,088	14,926	14,281	14,604	0.053088	0.014604	72.49 %
16-May-21	61,027	55,605	58,316	15,917	15,387	15,652	0.058316	0.015652	73.16 %
17-May-21	62,576	56,767	59,671	14,270	15,314	14,792	0.059671	0.014792	75.21 %
18-May-21	59,090	54,056	56,573	15,351	15,521	15,436	0.056573	0.015436	72.71 %

19-May-21	59,865	50,183	55,024	14,667	15,518	15,093	0.055024	0.015093	72.57 %
20-May-21	49,478	51,344	50,411	14,212	15,356	14,784	0.050411	0.014784	70.67 %
21-May-21	40,639	52,506	46,573	13,965	14,096	14,031	0.046573	0.014031	69.87 %
22-May-21	47,998	59,478	53,738	13,915	15,521	14,718	0.053738	0.014718	72.61 %
23-May-21	43,350	59,090	51,220	14,726	15,262	14,994	0.051220	0.014994	70.73 %
24-May-21	45,287	60,252	52,770	14,066	13,898	13,982	0.052770	0.013982	73.50 %
25-May-21	44,900	55,217	50,058	14,881	15,813	15,347	0.050058	0.015347	69.34 %
26-May-21	49,478	62,963	56,221	15,091	14,913	15,002	0.056221	0.015002	73.32 %
27-May-21	44,969	57,928	51,449	14,773	15,903	15,338	0.051449	0.015338	70.19 %
28-May-21	45,674	59,865	52,770	15,764	15,147	15,456	0.052770	0.015456	70.71 %
29-May-21	61,414	51,344	56,379	15,138	15,097	15,118	0.056379	0.015118	73.19 %
30-May-21	59,090	55,217	57,154	14,566	14,560	14,563	0.057154	0.014563	74.52 %
31-May-21	60,639	58,703	59,671	15,621	15,342	15,482	0.059671	0.015482	74.06 %
01-Jun-21	63,350	59,090	61,220	15,642	14,101	14,872	0.061220	0.014872	75.71 %
02-Jun-21	66,836	62,189	64,512	14,728	14,707	14,718	0.064512	0.014718	77.19 %

03-Jun-21	58,703	57,541	58,122	15,242	15,111	15,177	0.058122	0.015177	73.89 %
04-Jun-21	55,605	63,738	59,672	15,790	14,494	15,142	0.059672	0.015142	74.62 %
05-Jun-21	63,350	57,928	60,639	15,635	13,944	14,790	0.060639	0.014790	75.61 %
06-Jun-21	58,316	62,182	60,249	15,998	14,699	15,349	0.060249	0.015349	74.52 %
07-Jun-21	57,154	59,865	58,510	14,074	14,045	14,060	0.058510	0.014060	75.97 %
08-Jun-21	62,189	59,090	60,640	14,959	15,421	15,190	0.060640	0.015190	74.95 %
09-Jun-21	62,189	61,414	61,801	15,323	14,614	14,969	0.061801	0.014969	75.78 %
10-Jun-21	63,738	49,408	56,573	14,180	15,685	14,933	0.056573	0.014933	73.60 %
11-Jun-21	58,703	55,992	57,348	14,418	14,568	14,493	0.057348	0.014493	74.73 %
12-Jun-21	56,379	58,316	57,348	15,007	15,226	15,117	0.057348	0.015117	73.64 %
13-Jun-21	64,125	67,998	66,062	15,340	14,518	14,929	0.066062	0.014929	77.40 %
14-Jun-21	62,189	63,350	62,770	14,822	15,402	15,112	0.062770	0.015112	75.92 %
15-Jun-21	57,541	56,379	56,960	15,743	15,454	15,599	0.056960	0.015599	72.61 %
16-Jun-21	61,027	61,414	61,221	14,876	14,695	14,786	0.061221	0.014786	75.85 %
17-Jun-21	61,801	60,639	61,220	15,844	14,416	15,130	0.061220	0.015130	75.29 %
18-Jun-21	50,570	42,050	46,310	14,544	14,724	14,634	0.046310	0.014634	68.40 %
19-Jun-21	57,154	59,865	58,510	14,322	14,507	14,415	0.058510	0.014415	75.36 %
20-Jun-21	55,992	55,217	55,605	15,923	14,289	15,106	0.055605	0.015106	72.83 %
21-Jun-21	65,287	64,900	65,094	13,913	14,394	14,154	0.065094	0.014154	78.26 %
22-Jun-21	64,125	58,316	61,221	14,842	13,987	14,415	0.061221	0.014415	76.45 %
23-Jun-21	61,027	53,668	57,348	14,009	15,586	14,798	0.057348	0.014798	74.20 %
24-Jun-21	66,449	60,639	63,544	15,889	14,029	14,959	0.063544	0.014959	76.46 %

25-Jun-21	62,963	55,992	59,478	15,032	15,877	15,455	0.059478	0.015455	74.02 %
26-Jun-21	58,316	55,992	57,154	15,926	13,858	14,892	0.057154	0.014892	73.94 %
27-Jun-21	58,703	50,570	54,637	15,244	15,960	15,602	0.054637	0.015602	71.44 %
28-Jun-21	57,928	54,056	55,992	14,466	14,990	14,728	0.055992	0.014728	73.70 %
29-Jun-21	62,189	59,090	60,640	15,935	14,971	15,453	0.060640	0.015453	74.52 %
30-Jun-21	55,217	52,506	53,862	15,563	15,494	15,529	0.053862	0.015529	71.17 %
01-Jul-21	59,865	54,443	57,154	14,877	14,354	14,616	0.057154	0.014616	74.43 %
02-Jul-21	60,639	56,379	58,509	15,148	15,533	15,341	0.058509	0.015341	73.78 %
03-Jul-21	60,252	55,992	58,122	15,702	13,898	14,800	0.058122	0.014800	74.54 %
04-Jul-21	63,738	51,096	57,417	14,447	15,263	14,855	0.057417	0.014855	74.13 %
05-Jul-21	56,379	57,928	57,154	14,604	15,060	14,832	0.057154	0.014832	74.05 %
06-Jul-21	50,957	51,732	51,345	14,537	15,969	15,253	0.051345	0.015253	70.29 %
07-Jul-21	57,154	55,605	56,380	13,980	14,143	14,062	0.056380	0.014062	75.06 %
08-Jul-21	59,090	57,541	58,316	15,719	14,851	15,285	0.058316	0.015285	73.79 %
09-Jul-21	66,836	68,773	67,805	14,848	14,901	14,875	0.067805	0.014875	78.06 %
10-Jul-21	54,056	58,703	56,380	15,256	14,309	14,783	0.056380	0.014783	73.78 %
11-Jul-21	54,830	53,668	54,249	15,181	15,596	15,389	0.054249	0.015389	71.63 %
12-Jul-21	52,506	49,795	51,151	14,277	15,402	14,840	0.051151	0.014840	70.99 %
13-Jul-21	52,119	51,344	51,732	14,168	14,831	14,500	0.051732	0.014500	71.97 %
14-Jul-21	45,535	53,281	49,408	15,889	13,981	14,935	0.049408	0.014935	69.77 %
15-Jul-21	64,512	57,154	60,833	15,567	14,339	14,953	0.060833	0.014953	75.42 %
16-Jul-21	59,865	57,928	58,897	15,199	14,407	14,803	0.058897	0.014803	74.87 %

17-Jul-21	51,732	55,992	53,862	15,856	15,130	15,493	0.053862	0.015493	71.24 %
18-Jul-21	50,570	51,344	50,957	15,389	14,823	15,106	0.050957	0.015106	70.36 %
19-Jul-21	55,992	52,119	54,056	15,910	15,563	15,737	0.054056	0.015737	70.89 %
20-Jul-21	51,732	54,056	52,894	14,355	15,886	15,121	0.052894	0.015121	71.41 %
21-Jul-21	55,605	47,859	51,732	14,169	14,880	14,525	0.051732	0.014525	71.92 %
22-Jul-21	49,795	45,535	47,665	14,756	14,821	14,789	0.047665	0.014789	68.97 %
23-Jul-21	67,223	56,767	61,995	15,200	13,804	14,502	0.061995	0.014502	76.61 %
24-Jul-21	47,472	48,246	47,859	14,647	13,976	14,312	0.047859	0.014312	70.10 %
25-Jul-21	47,859	47,084	47,472	15,608	14,798	15,203	0.047472	0.015203	67.97 %
26-Jul-21	49,795	50,570	50,183	14,276	15,620	14,948	0.050183	0.014948	70.21 %
27-Jul-21	50,957	47,472	49,215	15,202	13,821	14,512	0.049215	0.014512	70.51 %
28-Jul-21	45,148	47,472	46,310	15,774	14,935	15,355	0.046310	0.015355	66.84 %
29-Jul-21	47,859	44,761	46,310	15,087	14,481	14,784	0.046310	0.014784	68.08 %
30-Jul-21	61,027	48,246	54,637	14,035	15,610	14,823	0.054637	0.014823	72.87 %
31-Jul-21	52,506	50,957	51,732	15,499	15,982	15,741	0.051732	0.015741	69.57 %
01-Aug-21	56,767	59,090	57,929	15,129	14,776	14,953	0.057929	0.014953	74.19 %
02-Aug-21	49,021	47,859	48,440	14,297	14,637	14,467	0.048440	0.014467	70.13 %
03-Aug-21	52,506	52,894	52,700	14,541	14,663	14,602	0.052700	0.014602	72.29 %
04-Aug-21	56,767	55,605	56,186	14,660	14,545	14,603	0.056186	0.014603	74.01 %
05-Aug-21	53,668	54,830	54,249	14,709	15,158	14,934	0.054249	0.014934	72.47 %
06-Aug-21	56,767	54,830	55,799	15,168	14,569	14,869	0.055799	0.014869	73.35 %
07-Aug-21	55,217	52,506	53,862	15,584	15,503	15,544	0.053862	0.015544	71.14 %

08-Aug-21	65,287	54,443	59,865	15,381	14,210	14,796	0.059865	0.014796	75.29 %
09-Aug-21	65,287	63,738	64,513	13,833	15,163	14,498	0.064513	0.014498	77.53 %
10-Aug-21	54,443	56,379	55,411	15,489	15,399	15,444	0.055411	0.015444	72.13 %
11-Aug-21	47,084	47,859	47,472	14,800	15,475	15,138	0.047472	0.015138	68.11 %
12-Aug-21	49,408	49,795	49,602	14,434	14,886	14,660	0.049602	0.014660	70.44 %
13-Aug-21	53,281	49,795	51,538	15,770	15,662	15,716	0.051538	0.015716	69.51 %
14-Aug-21	50,570	52,506	51,538	15,590	15,489	15,540	0.051538	0.015540	69.85 %
15-Aug-21	/	/	/	/	/	/	/	/	/
16-Aug-21	69,547	49,408	59,478	15,676	14,562	15,119	0.059478	0.015119	74.58 %
17-Aug-21	50,183	50,570	50,377	13,802	14,650	14,226	0.050377	0.014226	71.76 %
18-Aug-21	58,703	56,767	57,735	14,215	14,381	14,298	0.057735	0.014298	75.24 %
19-Aug-21	54,443	51,732	53,088	15,776	14,427	15,102	0.053088	0.015102	71.55 %
20-Aug-21	68,773	62,189	65,481	14,708	14,583	14,646	0.065481	0.014646	77.63 %
21-Aug-21	64,900	63,738	64,319	14,336	15,102	14,719	0.064319	0.014719	77.12 %
22-Aug-21	59,478	57,928	58,703	13,868	13,854	13,861	0.058703	0.013861	76.39 %
23-Aug-21	59,478	52,506	55,992	15,043	14,979	15,011	0.055992	0.015011	73.19 %
24-Aug-21	60,639	55,992	58,316	15,956	14,869	15,413	0.058316	0.015413	73.57 %
25-Aug-21	64,125	55,992	60,059	15,651	14,382	15,017	0.060059	0.015017	75.00 %
26-Aug-21	54,830	57,541	56,186	15,887	14,053	14,970	0.056186	0.014970	73.36 %
27-Aug-21	57,154	55,217	56,186	14,030	15,052	14,541	0.056186	0.014541	74.12 %
28-Aug-21	57,541	57,154	57,347	15,502	14,552	15,027	0.057347	0.015027	73.80 %
29-Aug-21	58,703	57,154	57,928	15,160	15,960	15,560	0.057928	0.015560	73.14 %

30-Aug-21	53,281	52,506	52,894	14,871	14,294	14,583	0.052894	0.014583	72.43 %
31-Aug-21	55,992	51,732	53,862	14,009	15,166	14,588	0.053862	0.014588	72.92 %
01-Sep-21	49,021	52,506	50,764	14,324	15,906	15,115	0.050764	0.015115	70.22 %
02-Sep-21	52,894	50,570	51,732	14,620	14,488	14,554	0.051732	0.014554	71.87 %
03-Sep-21	51,334	49,408	50,371	14,328	14,726	14,527	0.050371	0.014527	71.16 %
04-Sep-21	52,506	51,732	52,119	15,612	15,680	15,646	0.052119	0.015646	69.98 %
05-Sep-21	55,217	53,281	54,249	15,998	15,558	15,778	0.054249	0.015778	70.92 %
06-Sep-21	53,668	52,506	53,087	14,750	15,070	14,910	0.053087	0.014910	71.91 %
07-Sep-21	51,732	49,408	50,570	15,451	14,352	14,902	0.050570	0.014902	70.53 %
08-Sep-21	51,344	48,633	49,989	14,795	14,942	14,869	0.049989	0.014869	70.26 %
09-Sep-21	58,316	55,217	56,767	14,988	15,373	15,181	0.056767	0.015181	73.26 %
10-Sep-21	58,316	53,668	55,992	14,142	14,558	14,350	0.055992	0.014350	74.37 %
11-Sep-21	57,541	56,767	57,154	15,731	15,059	15,395	0.057154	0.015395	73.06 %
12-Sep-21	57,154	61,027	59,091	15,208	14,273	14,741	0.059091	0.014741	75.05 %
13-Sep-21	62,189	53,668	57,928	15,925	14,148	15,037	0.057928	0.015037	74.04 %
14-Sep-21	58,703	57,541	58,122	15,961	14,176	15,069	0.058122	0.015069	74.07 %
15-Sep-21	58,316	57,928	58,122	14,508	15,304	14,906	0.058122	0.014906	74.35 %
16-Sep-21	61,801	59,090	60,446	15,817	15,022	15,420	0.060446	0.015420	74.49 %
17-Sep-21	61,414	56,767	59,091	15,463	15,997	15,730	0.059091	0.015730	73.38 %
18-Sep-21	59,090	56,767	57,929	13,930	15,331	14,631	0.057929	0.014631	74.74 %
19-Sep-21	59,865	59,090	59,478	15,602	14,743	15,173	0.059478	0.015173	74.49 %
20-Sep-21	59,478	61,027	60,253	14,251	14,144	14,198	0.060253	0.014198	76.44 %

21-Sep-21	61,027	56,767	58,897	15,286	15,037	15,162	0.058897	0.015162	74.26 %
22-Sep-21	55,605	54,830	55,217	15,004	15,778	15,391	0.055217	0.015391	72.13 %
23-Sep-21	61,414	57,928	59,671	14,989	14,590	14,790	0.059671	0.014790	75.21 %
24-Sep-21	66,061	64,125	65,093	14,773	15,900	15,337	0.065093	0.015337	76.44 %
25-Sep-21	61,414	60,639	61,027	15,798	15,656	15,727	0.061027	0.015727	74.23 %
26-Sep-21	58,316	63,350	60,833	14,402	14,574	14,488	0.060833	0.014488	76.18 %
27-Sep-21	63,738	41,027	52,383	15,334	14,698	15,016	0.052383	0.015016	71.33 %
28-Sep-21	63,350	41,414	52,382	13,905	13,930	13,918	0.052382	0.013918	73.43 %
29-Sep-21	62,189	41,096	51,642	14,715	14,417	14,566	0.051642	0.014566	71.79 %
30-Sep-21	69,160	41,027	55,093	15,956	15,573	15,765	0.055093	0.015765	71.39 %
01-Oct-21	63,135	40,220	51,678	14,769	15,338	15,054	0.051678	0.015054	70.87 %
02-Oct-21	69,248	48,925	59,087	14,065	14,317	14,191	0.059087	0.014191	75.98 %
03-Oct-21	65,038	46,010	55,524	14,363	14,959	14,661	0.055524	0.014661	73.60 %
04-Oct-21	67,629	44,391	56,010	14,215	15,803	15,009	0.056010	0.015009	73.20 %
05-Oct-21	68,925	59,209	64,067	14,411	14,069	14,240	0.064067	0.014240	77.77 %
06-Oct-21	64,391	59,857	62,124	15,779	15,434	15,607	0.062124	0.015607	74.88 %
07-Oct-21	59,857	65,686	62,772	15,748	15,507	15,628	0.062772	0.015628	75.10 %
08-Oct-21	60,504	51,113	55,809	15,818	15,155	15,487	0.055809	0.015487	72.25 %
09-Oct-21	65,686	45,686	55,686	15,723	15,630	15,677	0.055686	0.015677	71.85 %
10-Oct-21	65,362	44,067	54,715	15,045	14,072	14,559	0.054715	0.014559	73.39 %
11-Oct-21	61,800	41,476	51,638	15,465	14,920	15,193	0.051638	0.015193	70.58 %
12-Oct-21	59,857	46,010	52,934	15,740	14,690	15,215	0.052934	0.015215	71.26 %

13- Oct-21	64,067	43,743	53,905	15,387	14,148	14,768	0.053905	0.014768	72.60 %
14- Oct-21	66,334	45,686	56,010	15,300	15,432	15,366	0.056010	0.015366	72.57 %
15- Oct-21	58,925	48,277	53,601	15,981	14,515	15,248	0.053601	0.015248	71.55 %
16- Oct-21	/	/	/	/	/	/	/	/	/
17- Oct-21	57,021	58,316	57,669	15,471	14,908	15,190	0.057669	0.015190	73.66 %
18- Oct-21	51,192	52,811	52,002	15,409	15,370	15,390	0.052002	0.015390	70.41 %
19- Oct-21	57,669	53,782	55,726	15,926	15,578	15,752	0.055726	0.015752	71.73 %
20- Oct-21	52,850	53,498	53,174	15,775	14,390	15,083	0.053174	0.015083	71.64 %
21- Oct-21	51,192	50,868	51,030	15,879	15,400	15,640	0.051030	0.015640	69.35 %
22- Oct-21	50,220	53,458	51,839	14,344	14,042	14,193	0.051839	0.014193	72.62 %
23- Oct-21	57,021	55,078	56,049	14,026	14,648	14,337	0.056049	0.014337	74.42 %
24- Oct-21	57,021	55,078	56,049	14,940	14,696	14,818	0.056049	0.014818	73.56 %
25- Oct-21	52,811	57,305	55,058	14,469	15,419	14,944	0.055058	0.014944	72.86 %
26- Oct-21	68,640	56,049	62,345	14,166	15,525	14,846	0.062345	0.014846	76.19 %
27- Oct-21	67,992	58,640	63,316	14,419	13,916	14,168	0.063316	0.014168	77.62 %
28- Oct-21	66,697	55,402	61,050	15,502	14,060	14,781	0.061050	0.014781	75.79 %
29- Oct-21	67,669	55,725	61,697	14,960	15,236	15,098	0.061697	0.015098	75.53 %
30- Oct-21	67,021	55,402	61,212	15,011	14,696	14,854	0.061212	0.014854	75.73 %
31- Oct-21	61,231	58,964	60,097	15,054	15,227	15,141	0.060097	0.015141	74.81 %
01- Nov-21	61,879	59,288	60,583	14,785	14,506	14,646	0.060583	0.014646	75.83 %
02- Nov-21	62,202	57,021	59,612	15,552	15,227	15,390	0.059612	0.015390	74.18 %
03- Nov-21	58,316	57,669	57,993	14,750	14,919	14,835	0.057993	0.014835	74.42 %

04-Nov-21	57,021	55,402	56,212	13,931	15,564	14,748	0.056212	0.014748	73.76 %
05-Nov-21	56,697	57,669	57,183	15,416	15,948	15,682	0.057183	0.015682	72.58 %
06-Nov-21	55,725	62,487	59,106	14,401	14,571	14,486	0.059106	0.014486	75.49 %
07-Nov-21	56,658	63,419	60,039	15,047	15,991	15,519	0.060039	0.015519	74.15 %
08-Nov-21	59,288	55,725	57,507	14,323	15,972	15,148	0.057507	0.015148	73.66 %
09-Nov-21	50,583	57,992	54,288	14,884	14,456	14,670	0.054288	0.014670	72.98 %
10-Nov-21	59,572	57,021	58,297	14,000	15,293	14,647	0.058297	0.014647	74.88 %
11-Nov-21	57,021	55,402	56,212	15,002	13,826	14,414	0.056212	0.014414	74.36 %
12-Nov-21	53,782	55,402	54,592	13,832	14,337	14,085	0.054592	0.014085	74.20 %
13-Nov-21	52,811	62,850	57,831	14,092	14,267	14,180	0.057831	0.014180	75.48 %
14-Nov-21	52,487	51,839	52,163	14,025	15,349	14,687	0.052163	0.014687	71.84 %
15-Nov-21	54,715	50,544	52,630	15,595	14,096	14,846	0.052630	0.014846	71.79 %
16-Nov-21	64,754	59,248	62,001	14,195	14,574	14,385	0.062001	0.014385	76.80 %
17-Nov-21	66,697	64,754	65,725	15,627	15,580	15,604	0.065725	0.015604	76.26 %
18-Nov-21	62,487	64,430	63,459	15,867	14,621	15,244	0.063459	0.015244	75.98 %
19-Nov-21	62,811	60,220	61,516	14,206	15,120	14,663	0.061516	0.014663	76.16 %
20-Nov-21	62,163	52,811	57,487	15,486	15,741	15,614	0.057487	0.015614	72.84 %
21-Nov-21	62,811	54,430	58,621	14,589	15,605	15,097	0.058621	0.015097	74.25 %
22-Nov-21	59,248	56,658	57,953	14,303	14,611	14,457	0.057953	0.014457	75.05 %
23-Nov-21	57,953	55,686	56,820	15,559	14,035	14,797	0.056820	0.014797	73.96 %
24-Nov-21	59,572	58,925	59,248	14,884	14,790	14,837	0.059248	0.014837	74.96 %
25-Nov-21	/	/	/	/	/	/	/	/	/

26- Nov-21	/	/	/	/	/	/	/	/	/
27- Nov-21	/	/	/	/	/	/	/	/	/
28- Nov-21	/	/	/	/	/	/	/	/	/
29- Nov-21	/	/	/	/	/	/	/	/	/
30- Nov-21	/	/	/	/	/	/	/	/	/
01- Dec-21	/	/	/	/	/	/	/	/	/
02- Dec-21	/	/	/	/	/	/	/	/	/
03- Dec-21	/	/	/	/	/	/	/	/	/
04- Dec-21	/	/	/	/	/	/	/	/	/
05- Dec-21	/	/	/	/	/	/	/	/	/
06- Dec-21	/	/	/	/	/	/	/	/	/
07- Dec-21	/	/	/	/	/	/	/	/	/
08- Dec-21	/	/	/	/	/	/	/	/	/
09- Dec-21	68,601	56,658	62,630	14,598	15,010	14,804	0.062630	0.014804	76.36 %
10- Dec-21	52,732	56,981	54,857	14,538	14,985	14,762	0.054857	0.014762	73.09 %
11- Dec-21	58,601	56,981	57,791	13,904	15,509	14,707	0.057791	0.014707	74.55 %
12- Dec-21	59,896	59,248	59,572	14,773	14,286	14,530	0.059572	0.014530	75.61 %
13- Dec-21	51,192	59,896	55,544	14,498	14,564	14,531	0.055544	0.014531	73.84 %
14- Dec-21	/	/	/	/	/	/	/	/	/
15- Dec-21	/	/	/	/	/	/	/	/	/
16- Dec-21	53,135	57,629	55,382	14,845	14,648	14,747	0.055382	0.014747	73.37 %
17- Dec-21	54,430	51,515	52,973	15,065	14,967	15,016	0.052973	0.015016	71.65 %

18-Dec-21	54,754	52,163	53,458	14,969	14,597	14,783	0.053458	0.014783	72.35 %
19-Dec-21	/	/	/	/	/	/	/	/	/
20-Dec-21	/	/	/	/	/	/	/	/	/
21-Dec-21	66,697	62,163	64,430	15,730	14,440	15,085	0.064430	0.015085	76.59 %
22-Dec-21	62,163	60,868	61,516	15,617	14,067	14,842	0.061516	0.014842	75.87 %
23-Dec-21	68,640	66,049	67,345	14,427	15,107	14,767	0.067345	0.014767	78.07 %
24-Dec-21	77,992	66,373	72,183	13,817	15,419	14,618	0.072183	0.014618	79.75 %
25-Dec-21	76,049	69,896	72,973	15,389	14,115	14,752	0.072973	0.014752	79.78 %
26-Dec-21	60,259	65,402	62,830	13,996	15,463	14,730	0.062830	0.014730	76.56 %
27-Dec-21	63,498	58,964	61,231	14,958	15,248	15,103	0.061231	0.015103	75.33 %
28-Dec-21	66,697	57,669	62,183	15,845	15,284	15,565	0.062183	0.015565	74.97 %
29-Dec-21	57,992	58,640	58,316	13,953	14,391	14,172	0.058316	0.014172	75.70 %
30-Dec-21	55,402	56,049	55,726	14,524	14,011	14,268	0.055726	0.014268	74.40 %
31-Dec-21	51,231	55,725	53,478	14,634	14,382	14,508	0.053478	0.014508	72.87 %
Average	58,381	55,206	56,805	14,920	14,930	14,924	0.056805	0.014924	76.33 %
Comment:	The treatment system is divided into the South and North zones, so the average value of South region and North region is taken as data result.								